St. George's University
Grenada, West Indies

Basic Principles of Medicine 1
Module: Musculoskeletal System | Lecture 11

Anterior Abdominal Wall & Inguinal Region

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Recommended Reading

- Drake, Vogl, Mitchell. Gray's anatomy for students. 5th ed. Elsevier.
- Printed Textbook: Chapter 4
- Online Textbook: Chapter 4
    - Regional Anatomy
        - Surface topography
        - Abdominal wall
        - Groin
- **Pay special attention to the blue "In the Clinic" boxes for the clinical correlations for this topic**

Your Objectives will show here!

Objectives

SOM.MKII.BPM1.1.MSK.1.ANAT.1242	Describe the bony boundaries of the anterolateral abdominal wall.
SOM.MKII.BPM1.1.MSK.1.ANAT.1243	List the dermatomes which overlie the xiphoid process, umbilicus and pubis.
SOM.MKII.BPM1.1.MSK.1.ANAT.1244	Describe the division of the anterolateral abdominal wall into four clinical quadrants and nine anatomical regions and its clinical significance.
SOM.MKII.BPM1.1.MSK.1.ANAT.1245	Describe the location of the major abdominal organs and vessels in the four quadrants and in the nine regions of the abdomen.
SOM.MKII.BPM1.1.MSK.1.ANAT.1246	Describe the blood supply, lymphatic drainage and cutaneous innervation of the abdominal wall.
SOM.MKII.BPM1.1.MSK.1.ANAT.1247	List the layers of the superficial abdominal fascia and describe their locations.
SOM.MKII.BPM1.1.MSK.1.ANAT.1248	List the general attachment, blood supply, innervations, and function of the muscles of the anterolateral abdominal wall.
SOM.MKII.BPM1.1.MSK.1.ANAT.1249	Describe how the aponeuroses of these muscles contribute to the anterior and posterior layers of the rectus sheath.
SOM.MKII.BPM1.1.MSK.1.ANAT.1250	Describe the inguinal ligament and its attachment sites.
SOM.MK.II.BPM1.1.MSK.1.ANAT.1505	Describe the boundaries and contents of the inguinal canal
SOM.MKII.BPM1.1.MSK.1.ANAT.1251	Describe the anatomy of the superficial and deep inguinal rings.
SOM.MKII.BPM1.1.MSK.1.ANAT.1252	Describe the boundaries of the inguinal triangle (of Hesselbach) and its clinical significance.
SOM.MKII.BPM1.1.MSK.1.ANAT.1253	Explain the difference between a direct and an indirect inguinal hernia.
SOM.MKII.BPM1.1.MSK.1.ANAT.1254	Explain why pain from an inguinal hernia occurs at the site of the hernia.
SOM.MKII.BPM1.1.MSK.1.ANAT.1255	Briefly describe testicular descent and its relationship to the anterior abdominal wall
SOM.MKII.BPM1.1.MSK.1.ANAT.1256	List the layers of spermatic fascia and where they arise from
SOM.MKII.BPM1.1.MSK.1.ANAT.1506	Describe the structures contained in the spermatic cord
SOM.MKII.BPM1.1.MSK.1.ANAT.1257	Describe the fate of the processus vaginalis and its contribution to an indirect inguinal hernia.
SOM.MKII.BPM1.1.MSK.1.ANAT.1258	Describe the formation of hydrocele
SOM.MKII.BPM1.1.MSK.1.ANAT.1259	Explain the surgical significance of the intersection of the right linea semilunaris and the costal margin.
SOM.MKII.BPM1.1.MSK.1.ANAT.1260	Explain the surgical significance of "McBurney's point".
SOM.MKII.BPM1.1.MSK.1.ANAT.1261	Explain the surgical approach through the layers of the abdominal wall to approach an inflamed appendix and the surgical significance of the muscle fiber orientation.
SOM.MKII.BPM1.1.MSK.1.ANAT.1262	Explain the advantage of performing vertical surgical incisions in the midline.

Lecture Overview

- **Abdominopelvic cavity & boundaries**
- **Surface topography & divisions** (quadrants, regions, landmarks)
- **Layers of the abdominal wall**
  - Fascia (Camper's, Scarpa's)
  - Anterolateral muscles
  - Rectus sheath
- **Neurovascular supply** (blood, lymphatics, innervation)
- **Inguinal region & Hasselbach's Triangle**
  - Inguinal ligament & canal
  - Spermatic cord & coverings
- **Inguinal hernias** (direct, indirect)
- **Surgical relevance**
  - McBurney's point
  - Abdominal incisions

Abdominopelvic cavity

The abdominal and pelvic cavities are continuous

Upper boundary

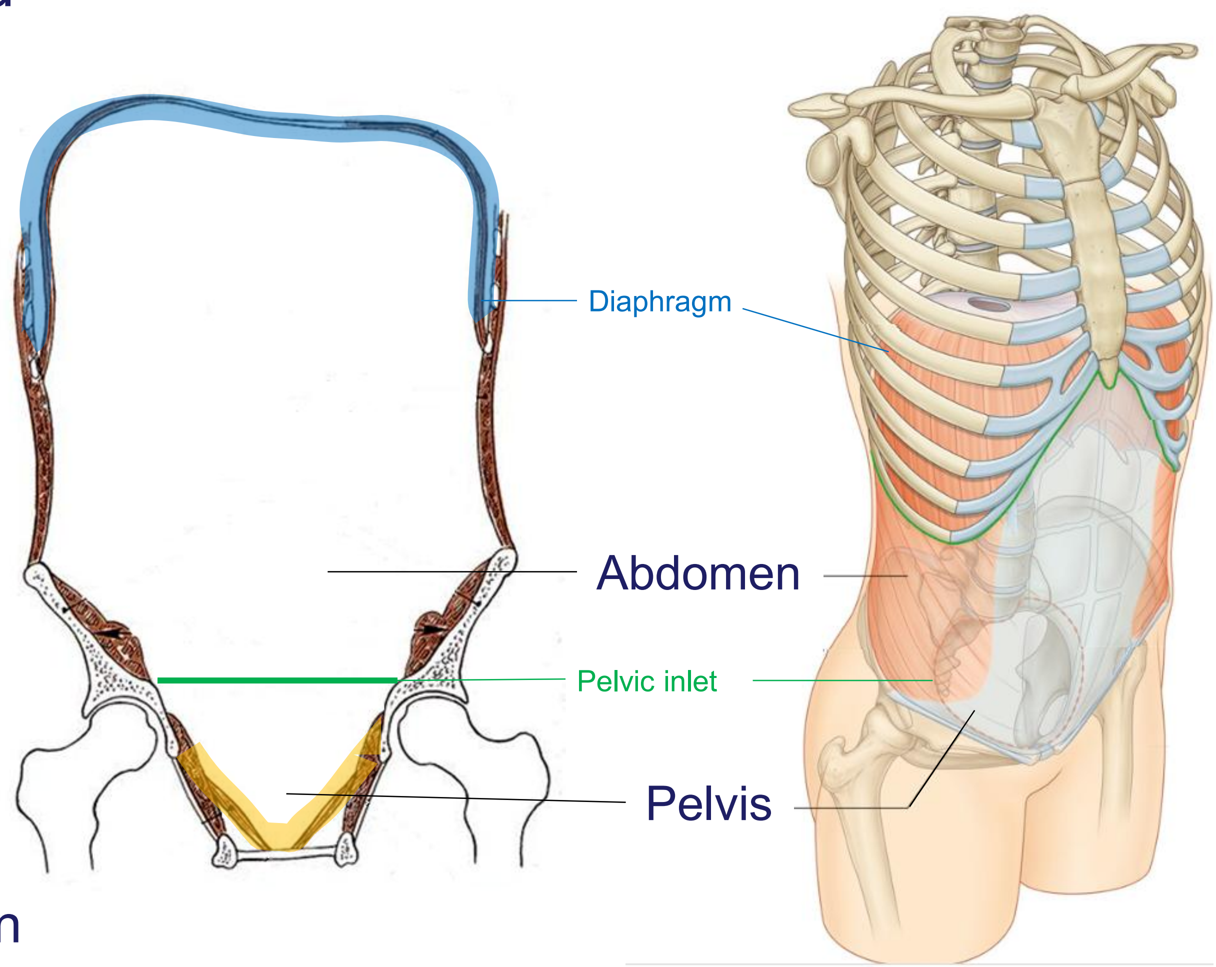
- Thoraco-abdominal diaphragm

Lower boundary

- pelvic diaphragm

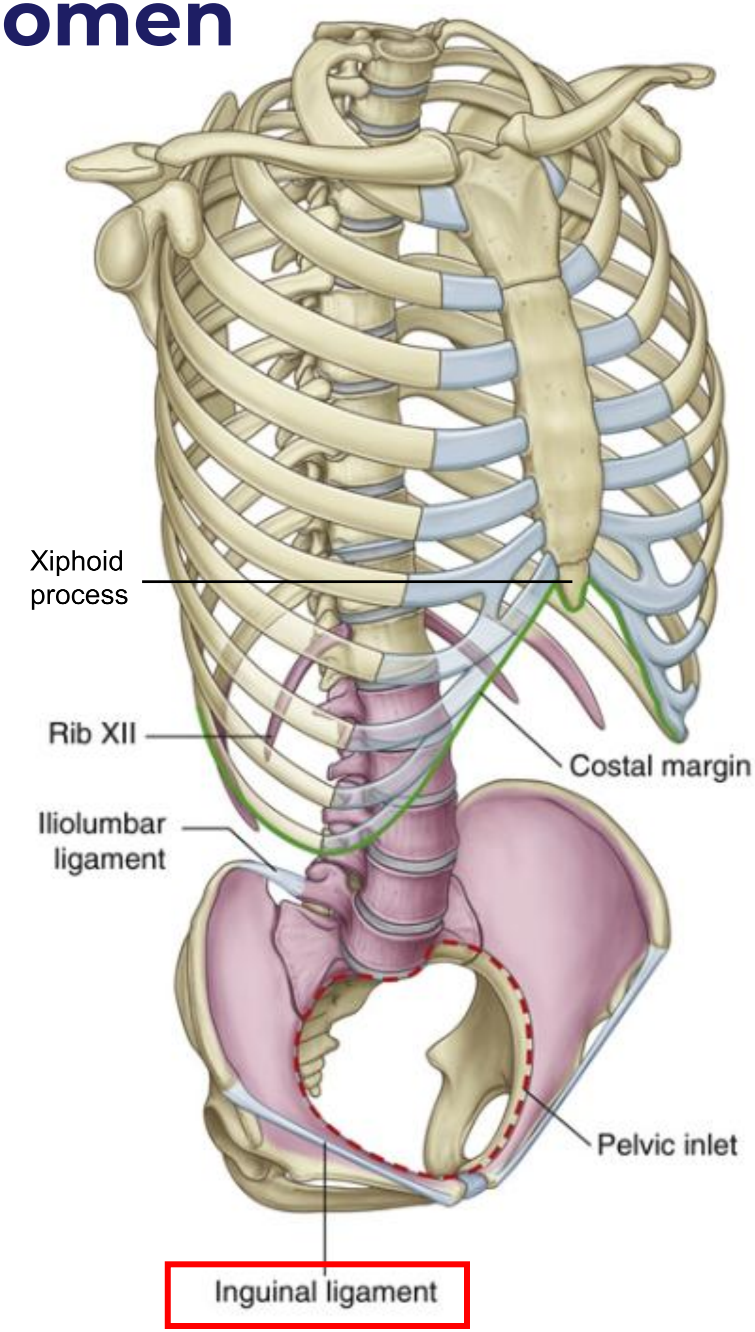
Pelvic inlet

- the division between the abdominal and pelvic cavities



Bony boundaries of abdomen

- Superior: xiphoid process & costal margins
- Posterior: vertebral column
- Inferior: upper parts of pelvic bones above the pelvic brim (pelvic inlet)

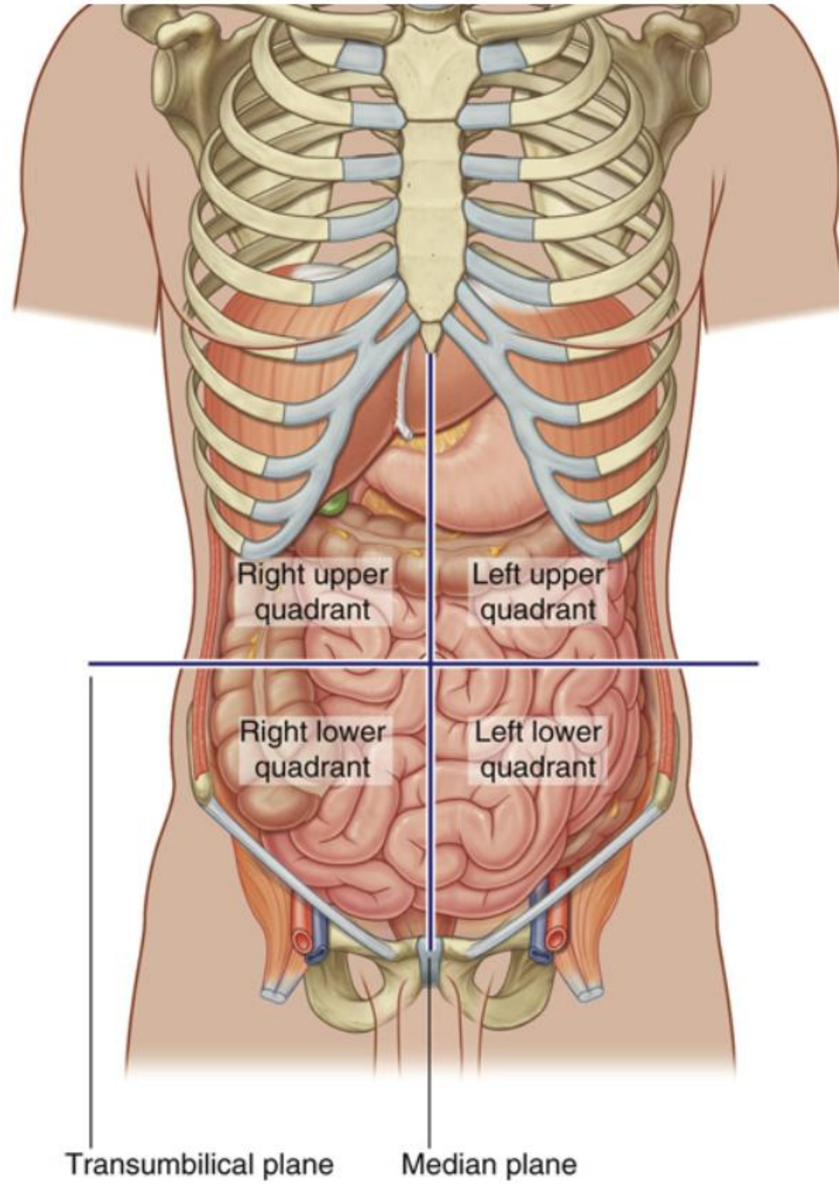


Abdomen: Four-quadrant pattern

- Abdomen can be divided into either **4 quadrants** or **9 regions** for physical examination and pain description.

- Planes used to divide abdomen into 4 quadrants are:
  - A vertical **median plane** (running through the midline).
  - A horizontal **transumbilical plane** (passing through the umbilicus and the L3–L4 intervertebral disc).

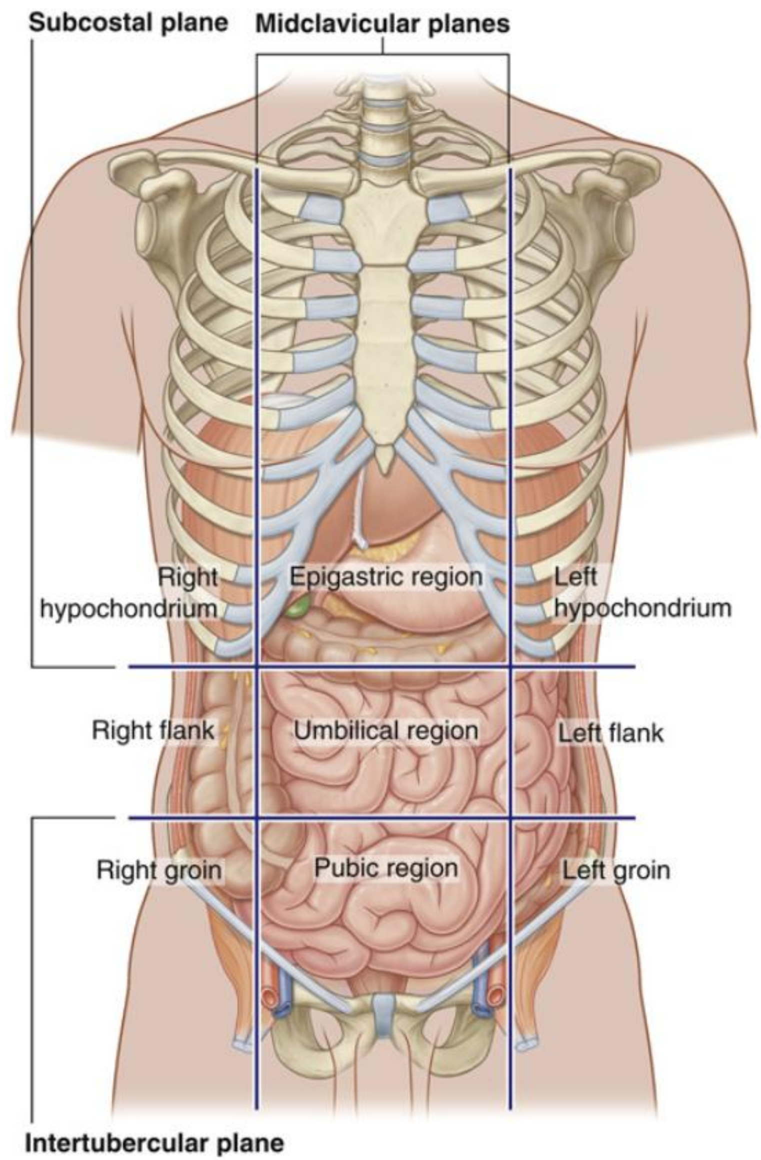
- Quadrants: **RUQ, LUQ, RLQ, LLQ**



Abdomen: Nine-region pattern

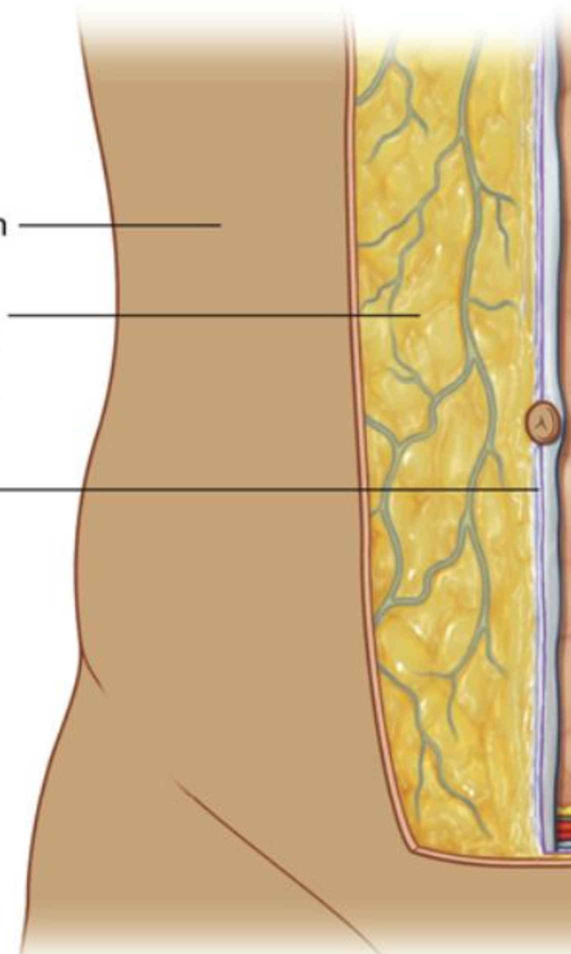
- Nine-region pattern: defined by 2 horizontal + 2 vertical planes.
- Horizontal planes:
  - Subcostal plane: just below costal margins; passes through L3.

    Sometimes transpyloric plane is used instead of subcostal which is halfway between umbilicus and the inferior end of sternum at L1
  - Intertubercular plane: connects iliac tubercles; passes through L5.
- Vertical planes: Midclavicular plane from midpoint of clavicles → midway between ASIS and pubic symphysis.
- Regions formed:
  - Superior row: Right hypochondrium – Epigastric – Left hypochondrium
  - Middle row: Right flank (lumbar) – Umbilical – Left flank (lumbar)
  - Inferior row: Right groin (inguinal) (iliac) – Pubic (Hypogastric) – Left groin (inguinal) (iliac)



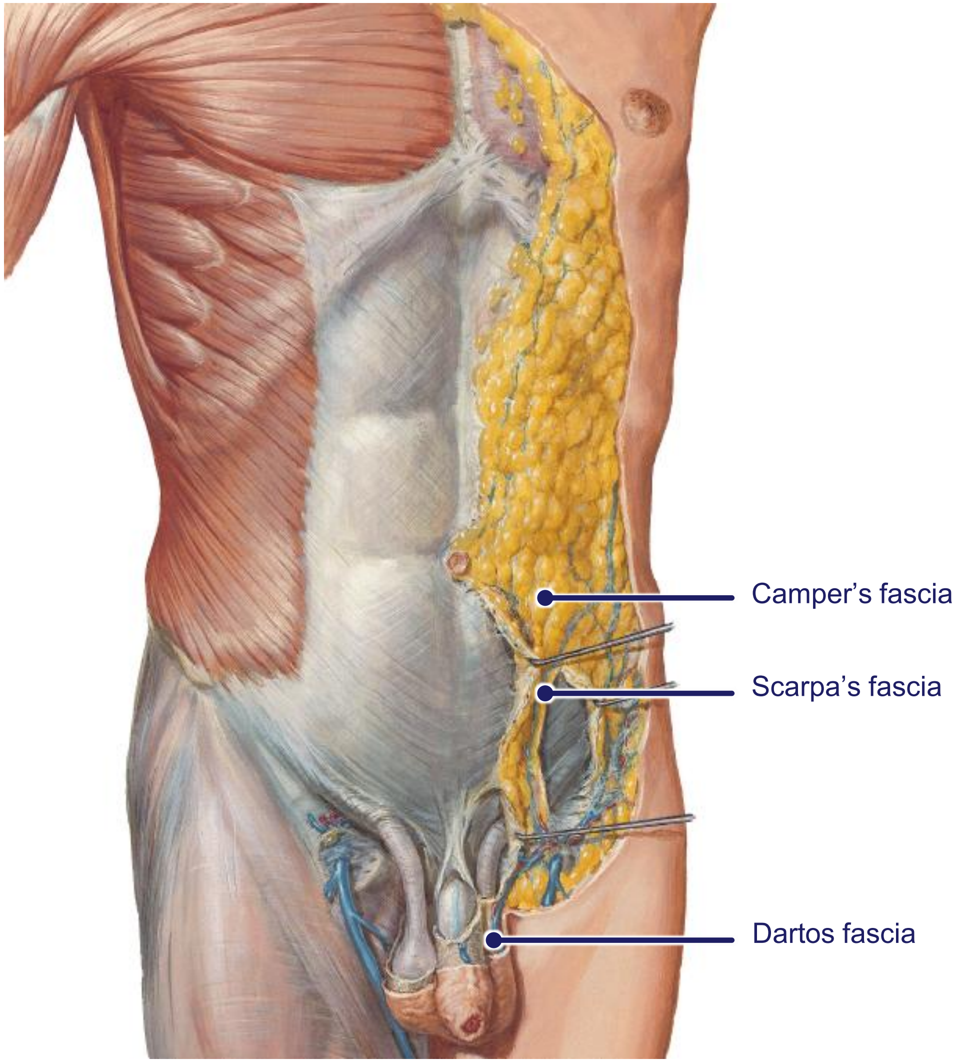
Abdominal fascia

- Superficial fascia of the abdominal wall (subcutaneous tissue of the abdomen) is a layer of fatty connective tissue.
- It is usually a single layer similar to, and continuous with, the superficial fascia throughout other regions of the body (thorax and perineum)
- Below the umbilicus - it forms two layers:
  - **Superficial Fatty Layer (Camper's fascia)**
  - **Deeper Membranous Layer (Scarpa's fascia)**



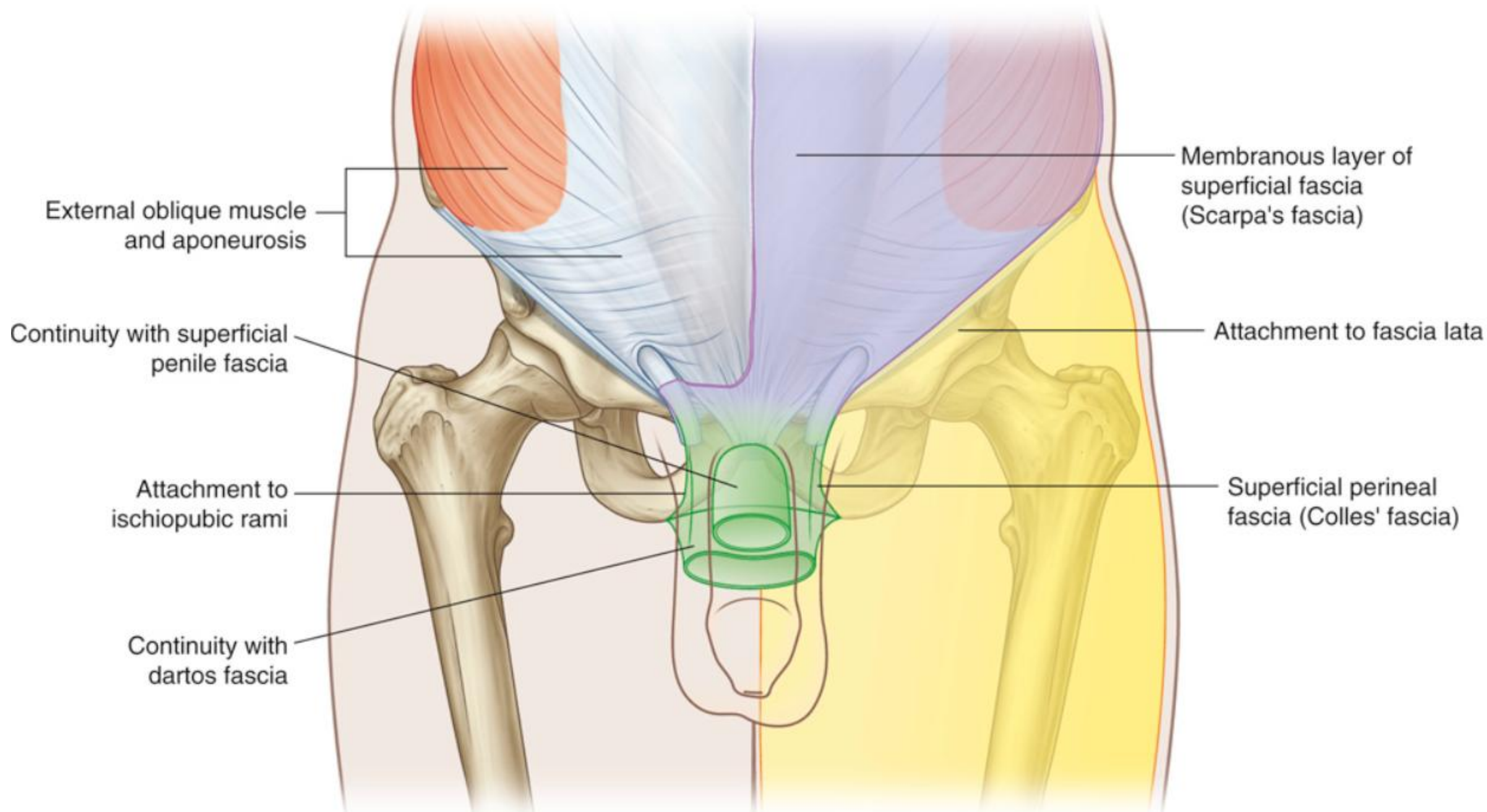
Camper's fascia

- Contains fat; thickness varies
- Continuous with superficial fascia of thigh & perineum
  - **Men**: loses most of its fat, fuses with Scarpa's layer to form Dartos fascia
  - **Women**: retains the fat, forms part of labia majora



Scarpa's fascia

- **Deeper Membranous** Layer that is **thin**, with **little/no fat**
- Continues into the thigh and fuses with **fascia lata (deep fascia of thigh)**
- Attaches firmly on the midline with **linea alba** & **pubic symphysis**
- Continues into the anterior part of perineum and attached into ischiopubic rami & perineal membrane (In perineum, it becomes **Colles' fascia (superficial fascia of perineum)**



Muscles of Anterolateral Abdominal wall

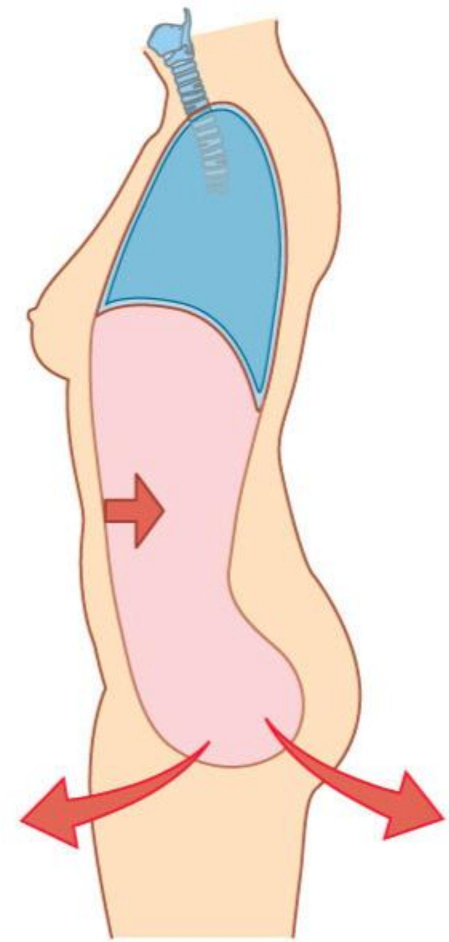
- Five muscles:
  - **Flat (3 muscles)**: **External oblique, Internal oblique, Transversus abdominis**

    Originate posterolaterally → Course anteriorly → End as aponeuroses

  - **Vertical (2 muscles): Rectus abdominis, Pyramidalis**

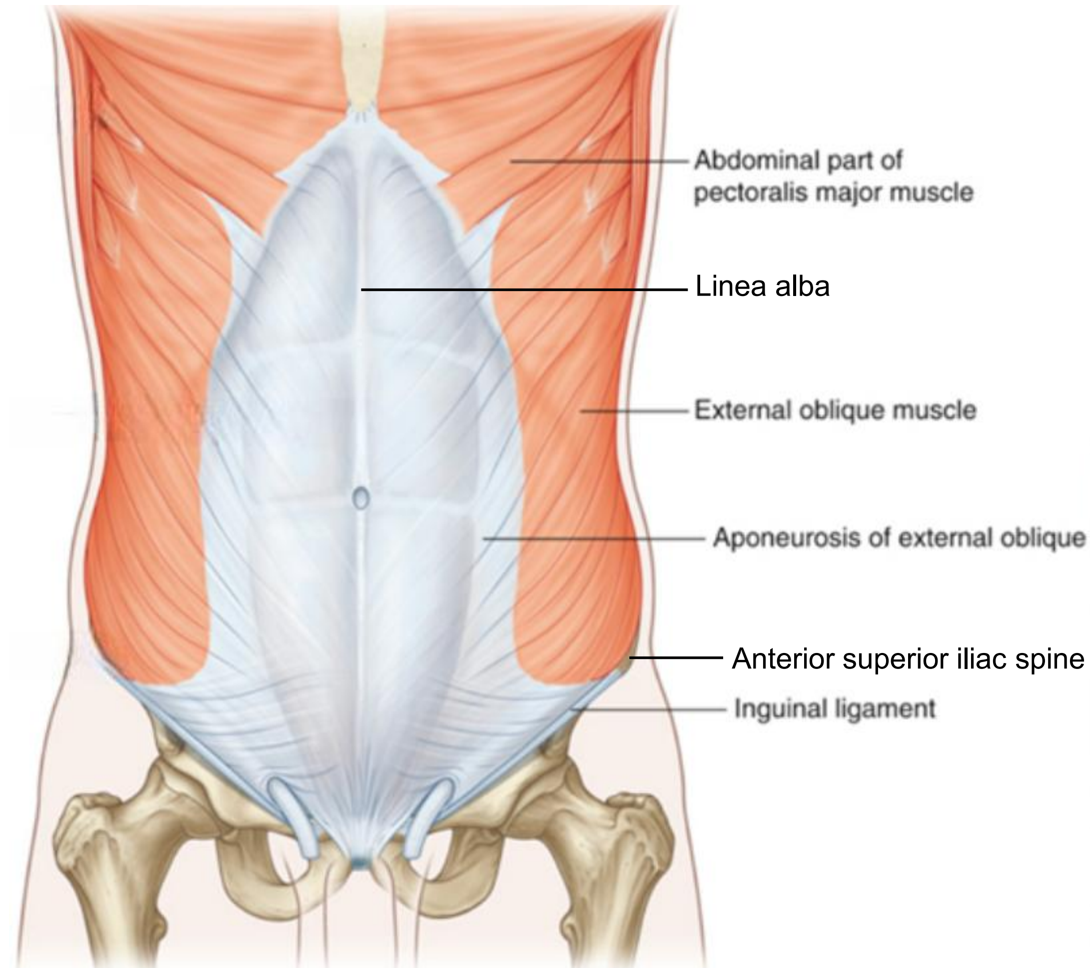
    Near midline

- Functions (together):
  - Form strong but flexible abdominal wall
  - Protect & support abdominal viscera
  - Maintain viscera position against gravity
  - Aid respiration (forced expiration), coughing, vomiting
  - Increase intra-abdominal pressure (childbirth, urination, defecation)



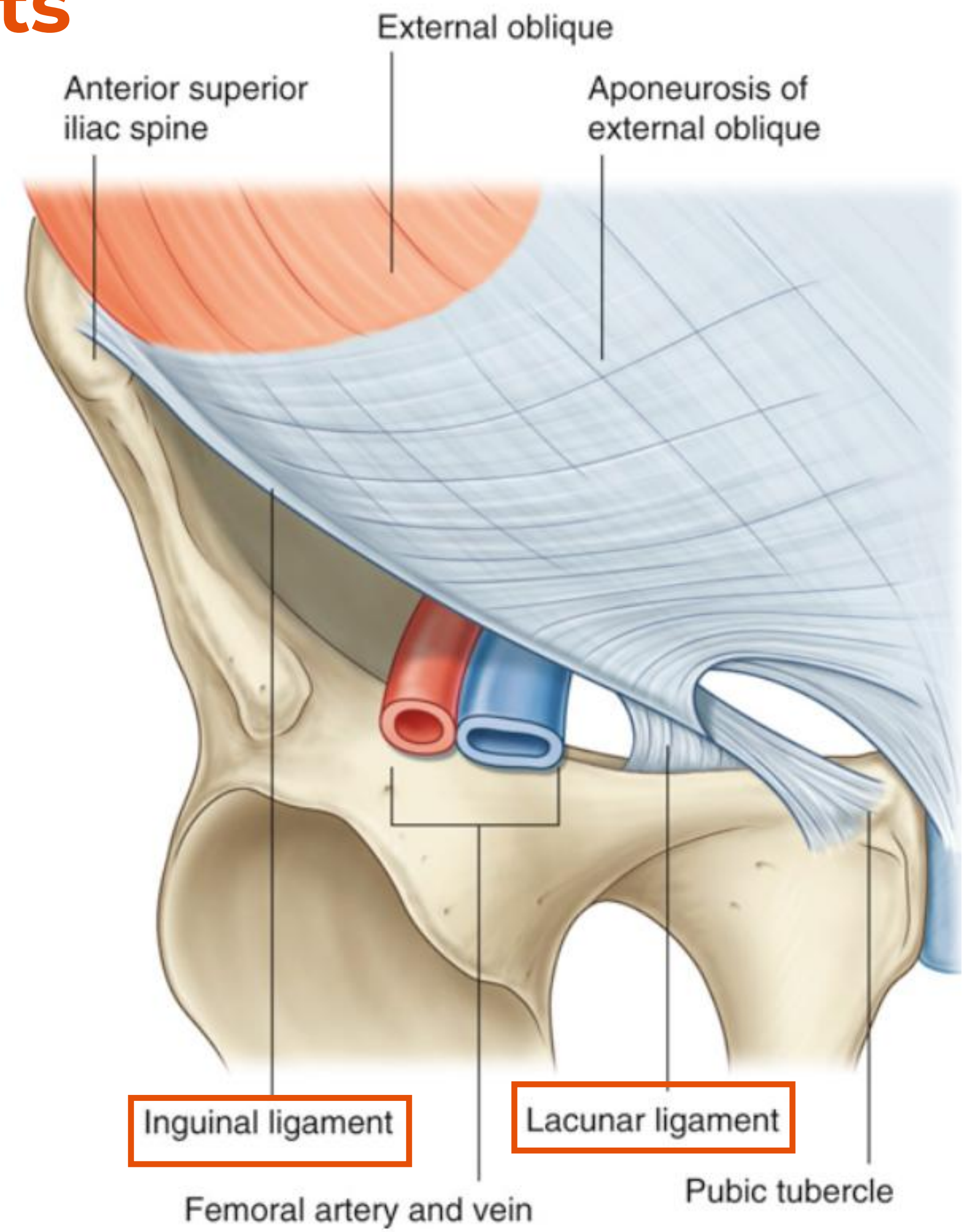
Muscles of the lateral wall: External oblique (EO)

- Most superficial; fibers run **downward-medially**
- Aponeurosis forms **linea alba** at midline
- Lower border forms **inguinal ligament**
- Innervated by segmental nerves (thoracoabdominal T7-12)



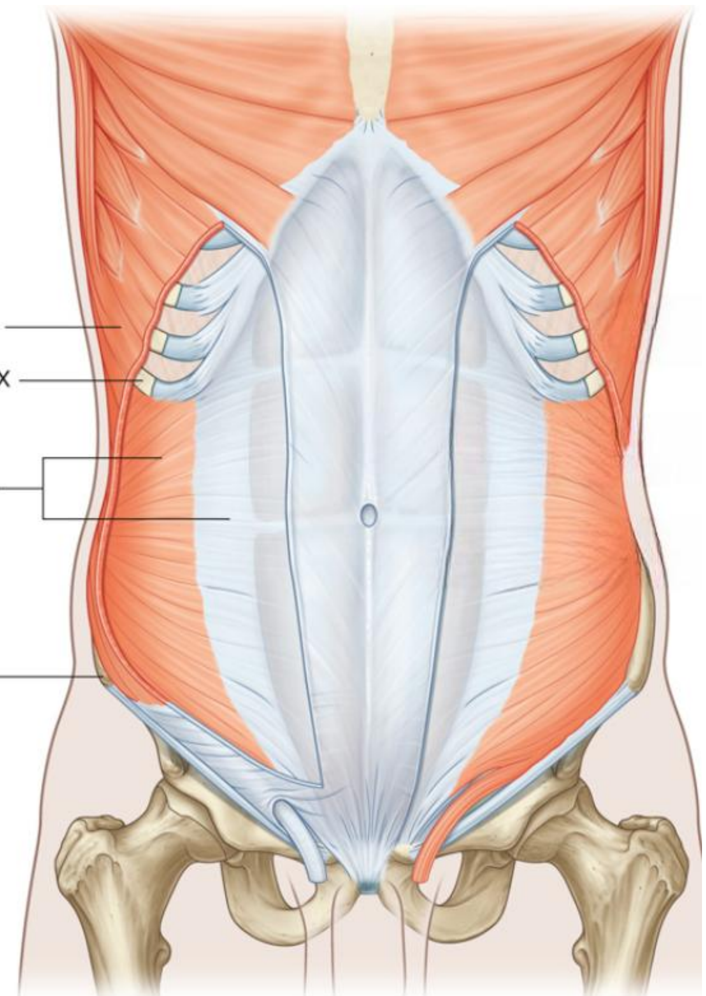
External oblique (EO): Associated ligaments

- **Inguinal ligament**
  - Formed from the lower border of the external oblique aponeurosis.
  - Extends from the **anterior superior iliac spine (ASIS)** laterally → **pubic tubercle** medially.
  - Plays a key role in the formation of the **inguinal canal.**
- **Lacunar ligament**
  - A **crescent-shaped extension** at the medial end of the inguinal ligament.



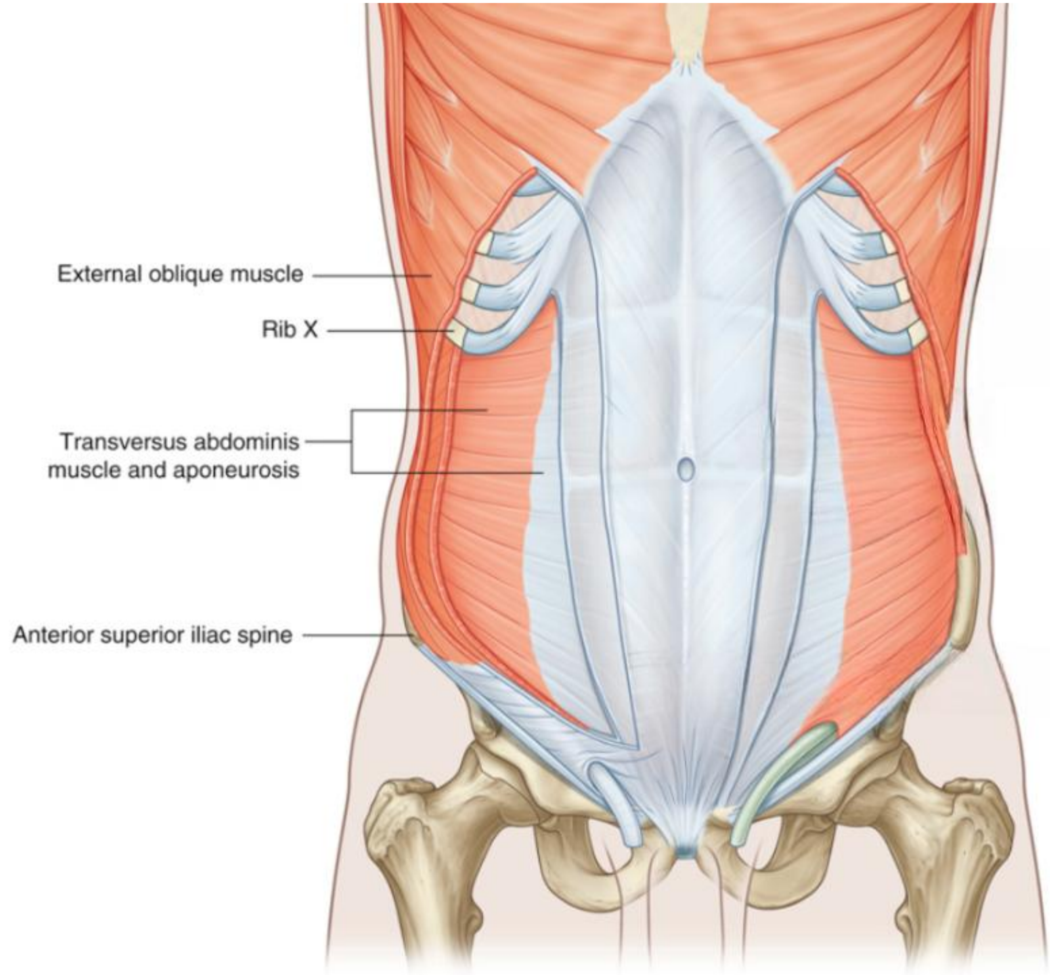
Muscles of the lateral wall: Internal oblique (IO)

- Deep to external oblique, thinner/smaller
- Most fibers run **upward-medially**, at right angles to external oblique
- Inferior portion joins with aponeurosis of transversus abdominis to form conjoint tendon
- Innervated by segmental nerves (thoracoabdominal T7-12, L1)



Muscles of the lateral wall: Transversus abdominus (TA)

- Deepest flat muscle; fibers run **transversely**
- Inferior portion joins with internal oblique to form conjoint tendon
- Deep surface lined by transversalis fascia
- Innervated by segmental nerves (thoracoabdominal T7-12, L1)

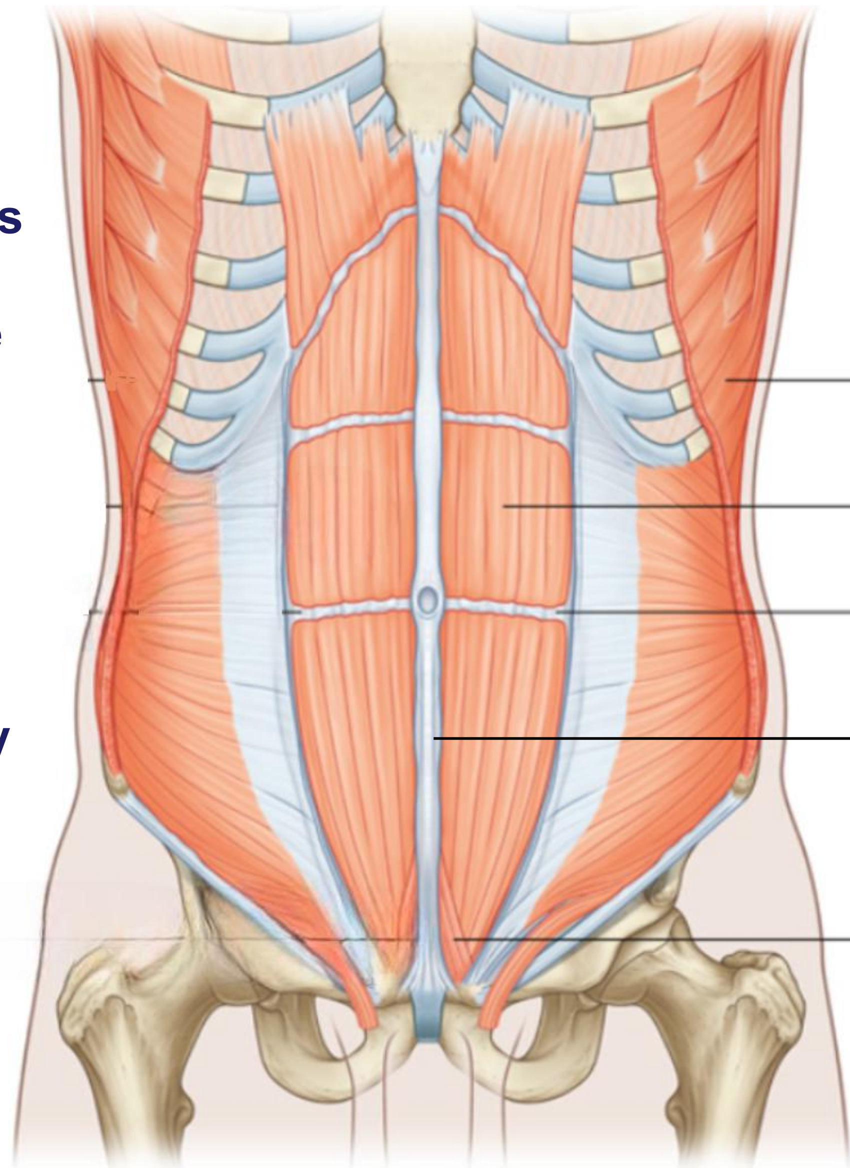


Muscles of the Anterior wall: Rectus abdominis

- Long, flat, paired muscle with **straight** fibers.
- Extends from **pubic symphysis** → **costal margin**.
- Separated at the midline by the **linea alba**.
- Interrupted by **3–4 transverse tendinous intersections** (visible in well-developed muscles as the "six-pack").

Pyramidalis

- Small, **triangular muscle** (**may be absent**).
- Lies anterior to rectus abdominis.
- Base: **pubis**
- Apex: **linea alba** (superior and medial attachment).



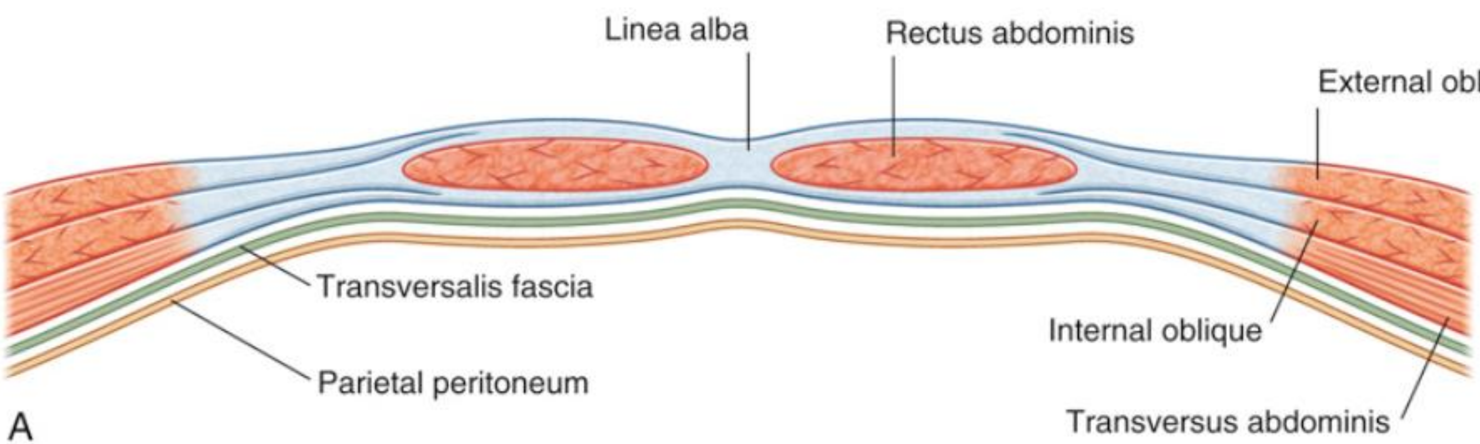
Summary Table Anterolateral Abdominal wall

MUSCLE	FUNCTION	INNERVATION
External oblique	Compress abdominal contents; both muscles flex trunk; each muscle bends trunk to same side, turning anterior part of abdomen to opposite side	Anterior rami of lower six thoracic spinal nerves (T7 to T12)
Internal oblique	Compress abdominal contents; both muscles flex trunk; each muscle bends trunk and turns anterior part of abdomen to same side	Anterior rami of lower six thoracic spinal nerves (T7 to T12) and L1
Transversus abdominis	Compress abdominal contents	Anterior rami of lower six thoracic spinal nerves (T7 to T12) and L1
Rectus abdominis	Compress abdominal contents; flex vertebral column; tense abdominal wall	Anterior rami mainly of the lower six thoracic spinal nerves (T7 to T12)
Pyramidalis	Tenses the linea alba	Anterior ramus of T12

Rectus Sheath

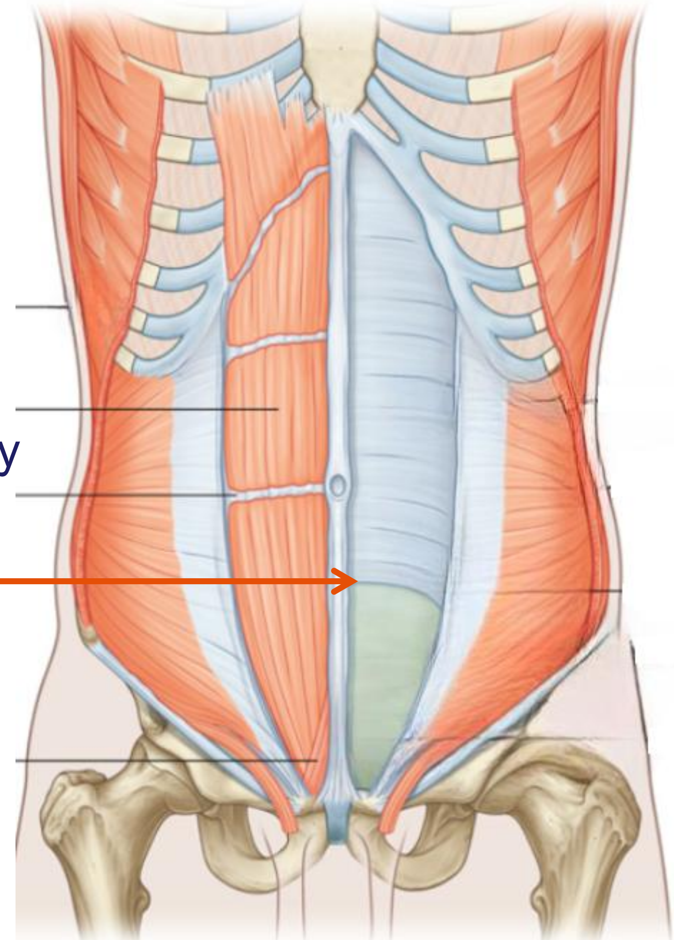
- Aponeurotic sheath enclosing **rectus abdominis** and **pyramidalis**, formed by aponeuroses of external oblique, internal oblique, and transversus abdominis.

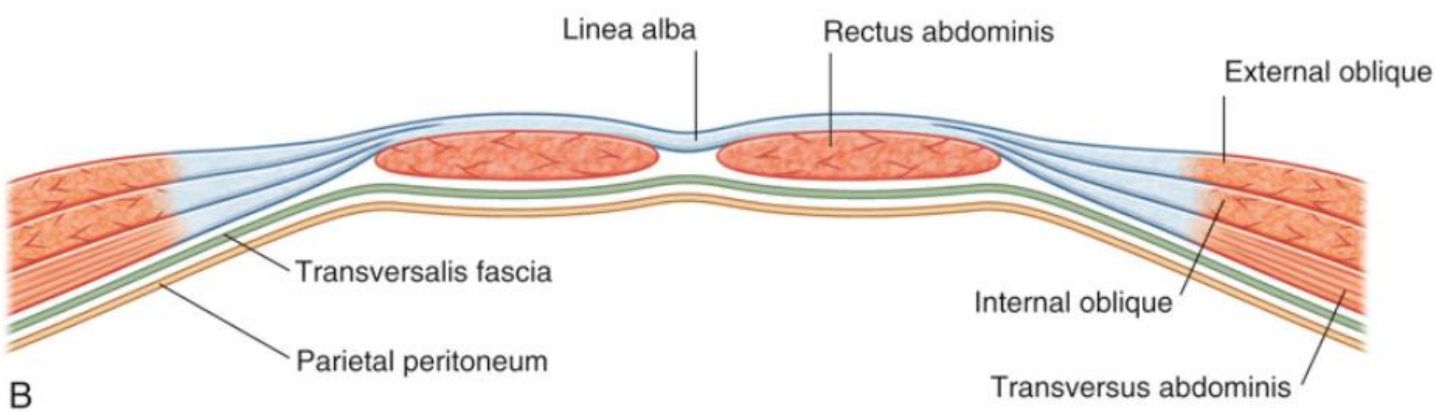
- **Upper ¾ of rectus abdominis** → completely enclosed by the sheath.
  - **Anterior wall**: external oblique aponeurosis + **½ of internal oblique aponeurosis** (splits at lateral margin).
  - **Posterior wall**: other ½ of internal oblique aponeurosis + transversus abdominis aponeurosis.



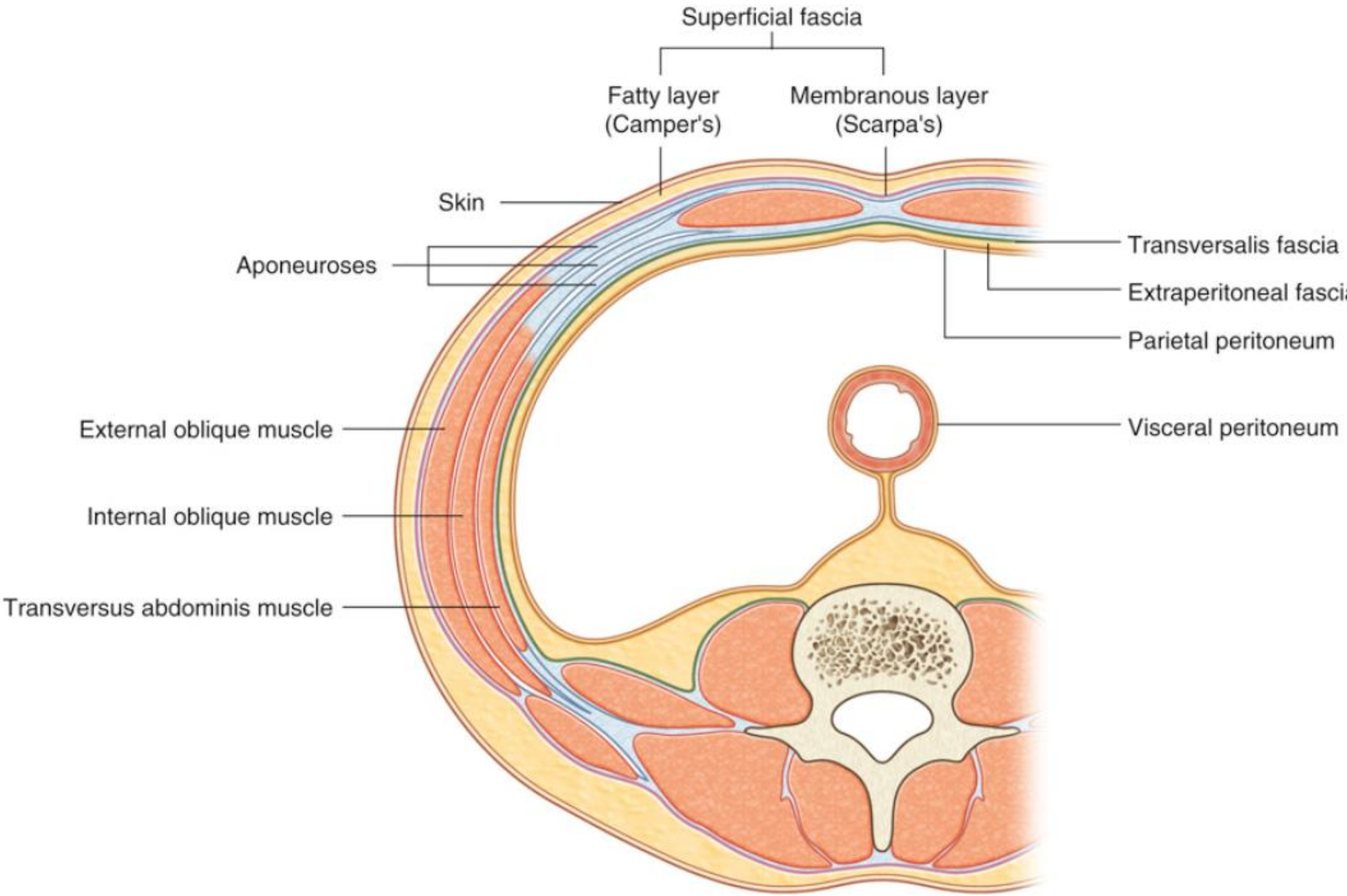
Rectus Sheath

- **Lower ¼ of rectus abdominis** → only anteriorly covered.
  - All three aponeuroses (external, internal, transversus) pass **anterior to rectus abdominis**.
  - No posterior wall → rectus abdominis directly contacts **transversalis fascia**.
- **Arcuate line:**
  - Transition point between upper ¾ and lower ¼.
  - Located midway between **umbilicus** and **pubic symphysis**.
  - Marked by an arch of fibers on posterior surface.





Transverse Section of the Abdominal Wall



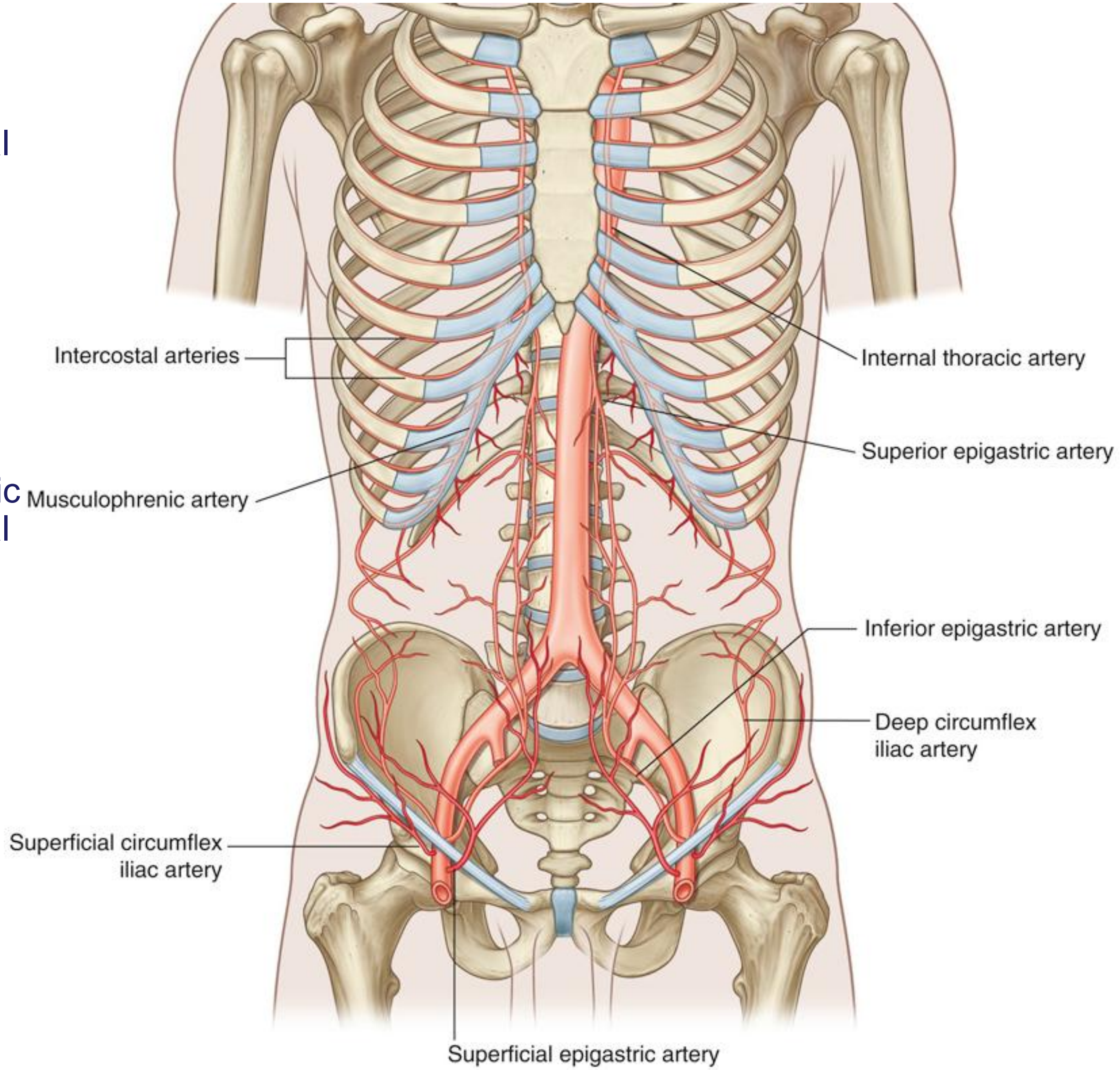
Blood supply

Superficial supply

- **Superior part**: musculophrenic artery (terminal branch of internal thoracic artery)
- **Inferior part**: superficial epigastric artery + superficial circumflex iliac artery (both branches of the femoral artery)

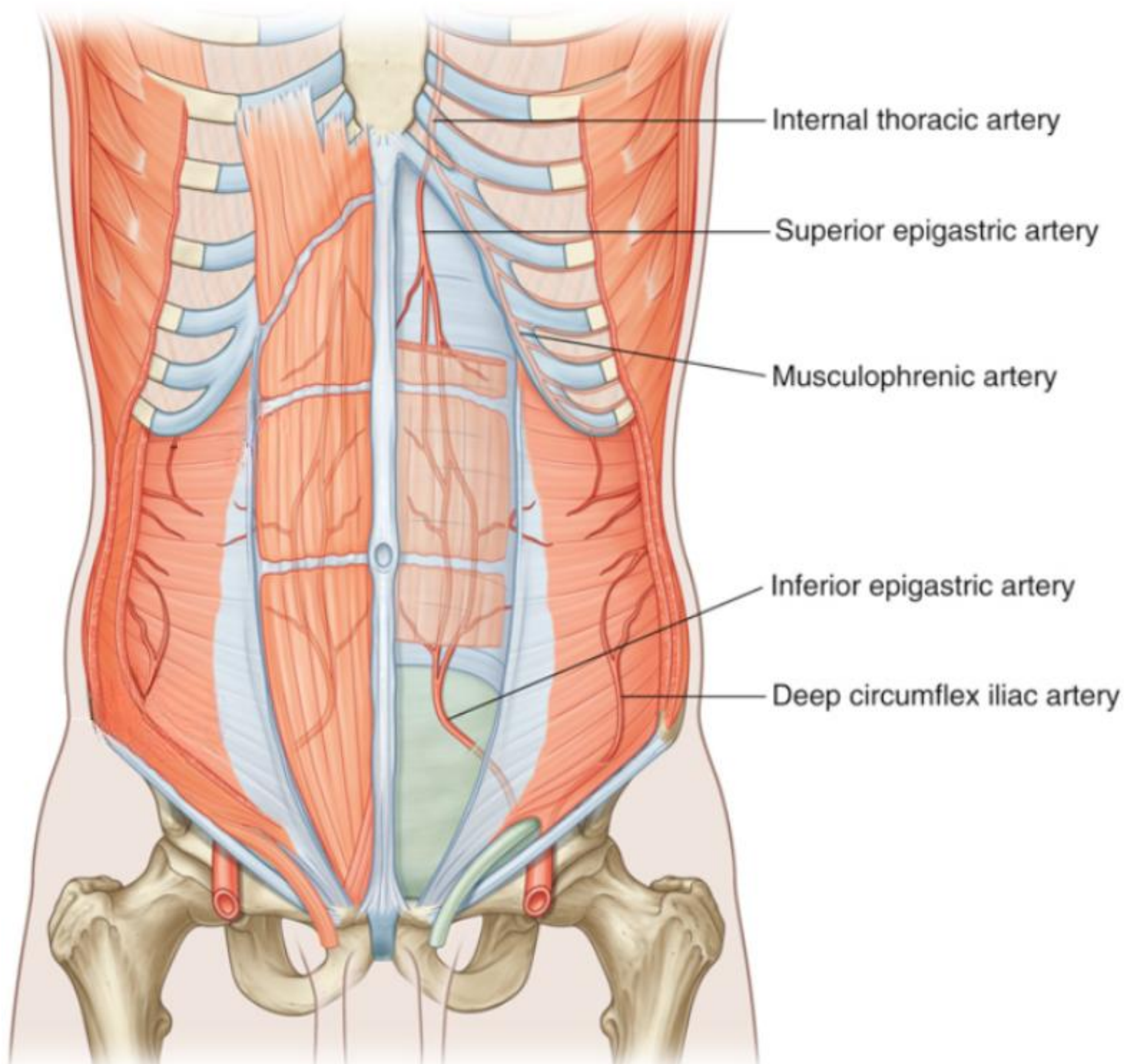
Deep supply

- **Superior part**: superior epigastric artery (terminal branch of internal thoracic artery)
- **Lateral part**: branches of 10th & 11th intercostal arteries + subcostal artery
- **Inferior part**: inferior epigastric artery + deep circumflex iliac artery (both branches of the external iliac artery)



Blood supply

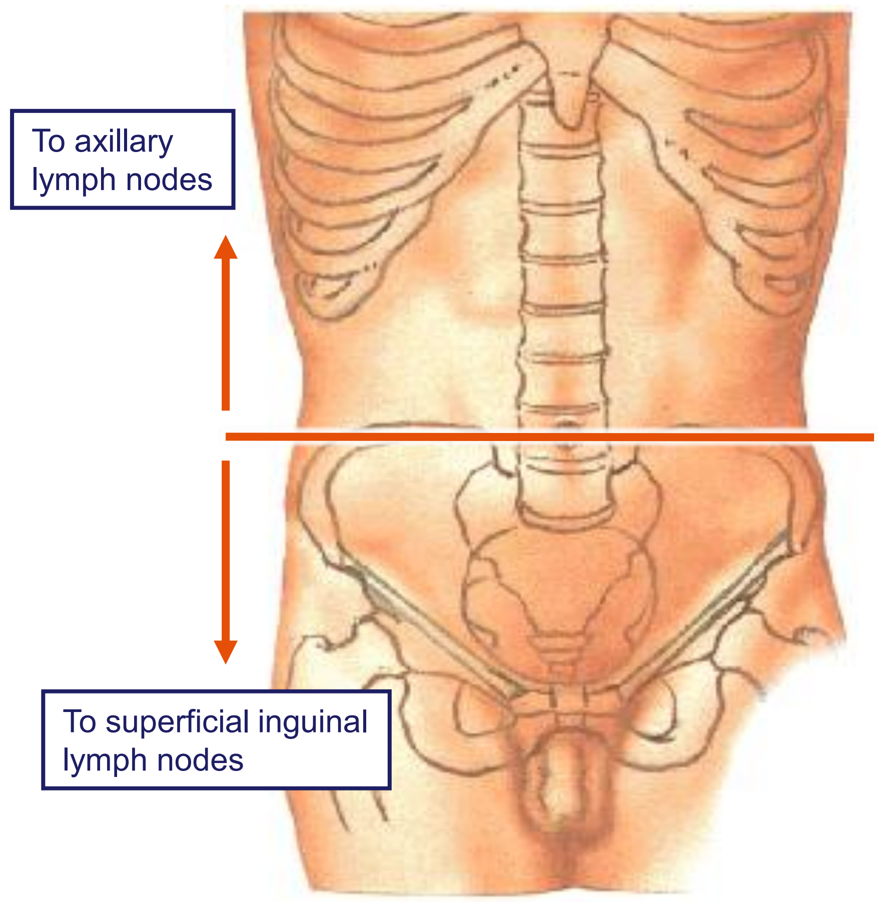
- **Special notes:** Superior and inferior epigastric arteries:
  - Enter **rectus sheath**
  - Run posterior to rectus abdominis
  - **Anastomose** with each other



- **Venous Drainage**
- Veins accompany arteries with **similar names** and provide venous return.

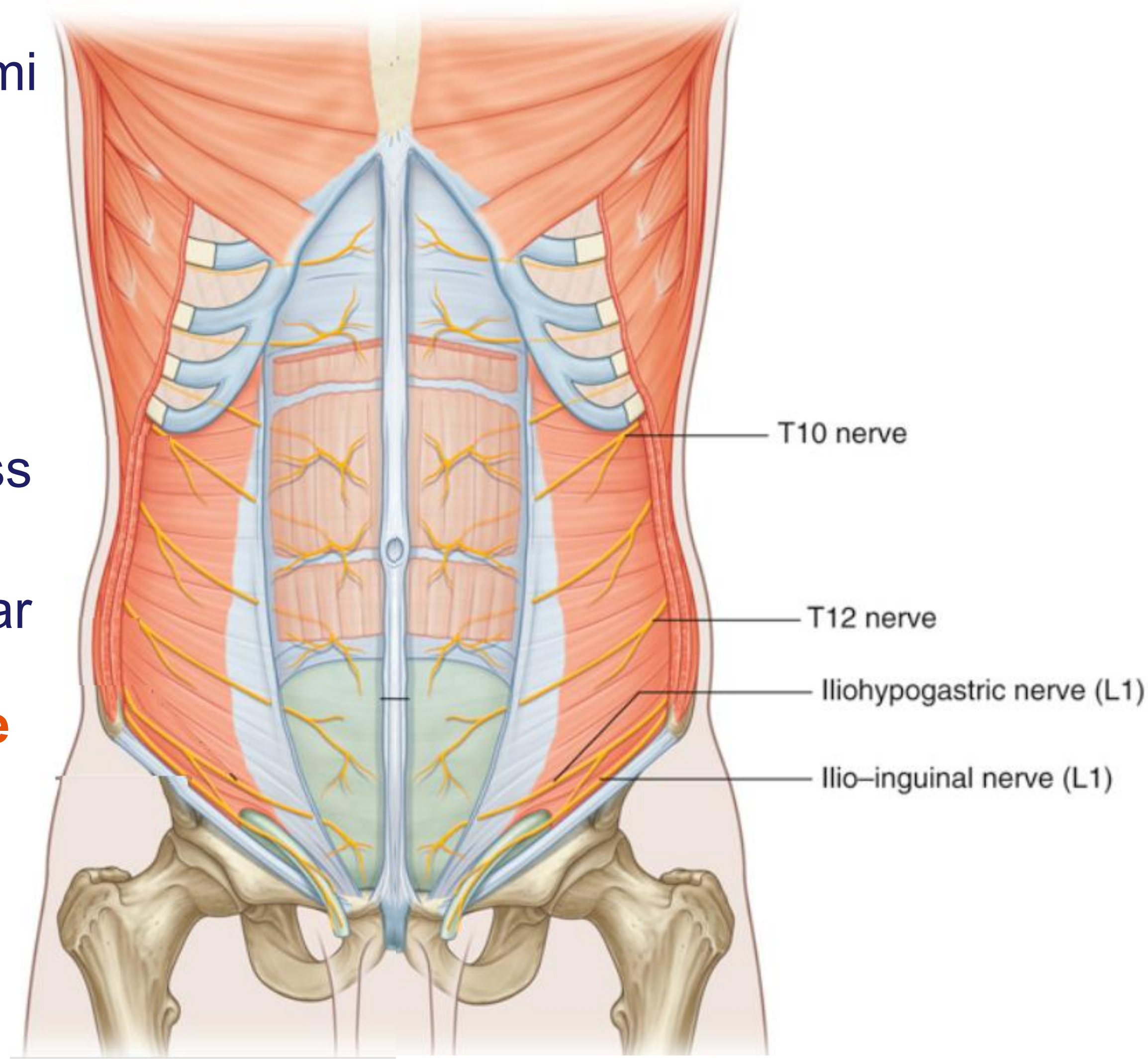
Lymphatic drainage

- **Superficial lymphatics**
  - **Above umbilicus** → drain superiorly to **axillary nodes**
  - **Below umbilicus** → drain inferiorly to **superficial inguinal nodes**
- **Deep lymphatics** (follow arteries)



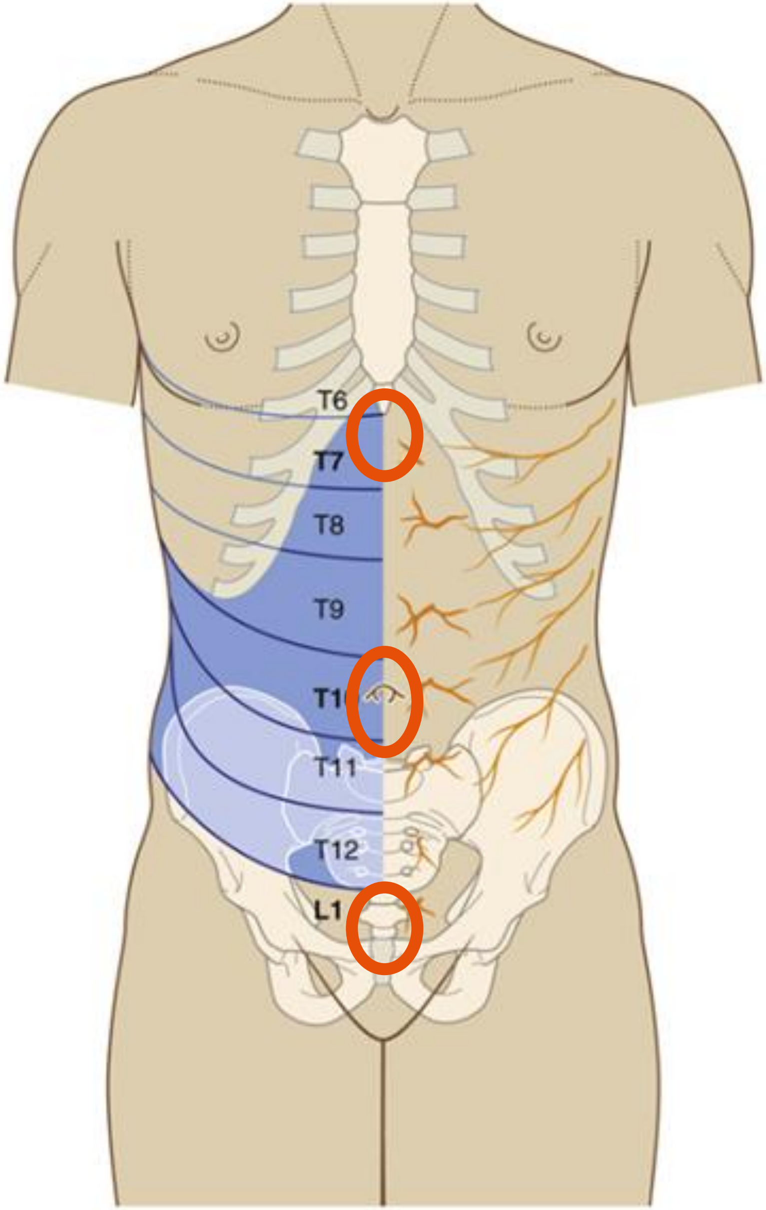
Innervation of Anterolateral Abdominal Wall

- **Main nerves**: anterior rami of **T7–T12** and **L1** spinal nerves .
- Run between **internal oblique & transversus abdominis**.
- Enter **rectus sheath**, pass behind rectus abdominis.
- **L1 branches** (from lumbar plexus): **Iliohypogastric nerve, Ilioinguinal nerve**



Innervation of Anterolateral Abdominal Wall

- **Innervation is segmental**
  - Motor supply to muscles = Somatic **e**fferent
  - Cutaneous sensory supply to skin (dermatomes) = Somatic **a**fferent
- **Important dermatomes**:
  - **T7** → xiphoid
  - **T10** → umbilicus
  - **L1** → Symphysis pubis

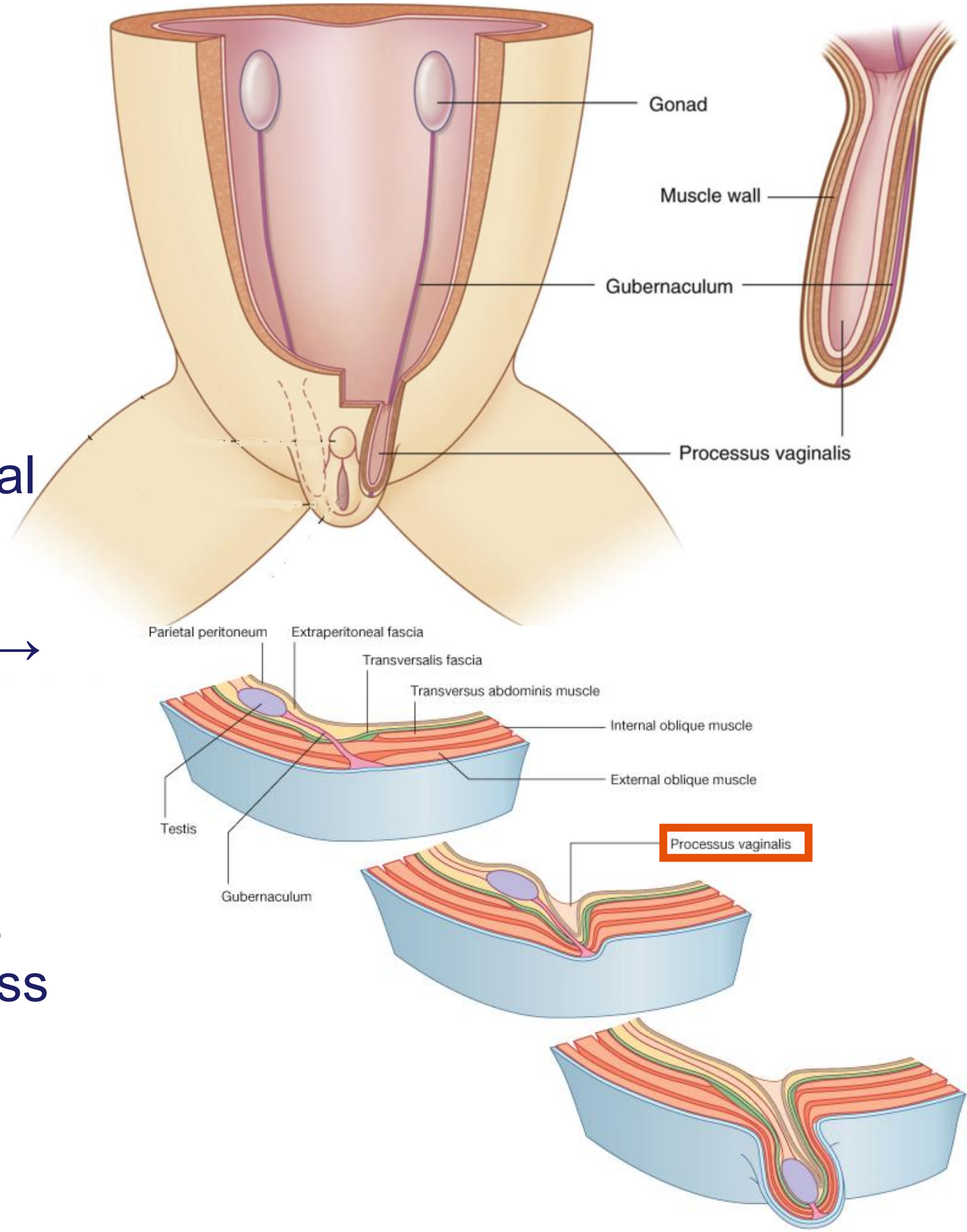


Inguinal region (Groin)

- Junction between **anterior abdominal wall** and **thigh**.
- Naturally weak site due to developmental descent of the gonads

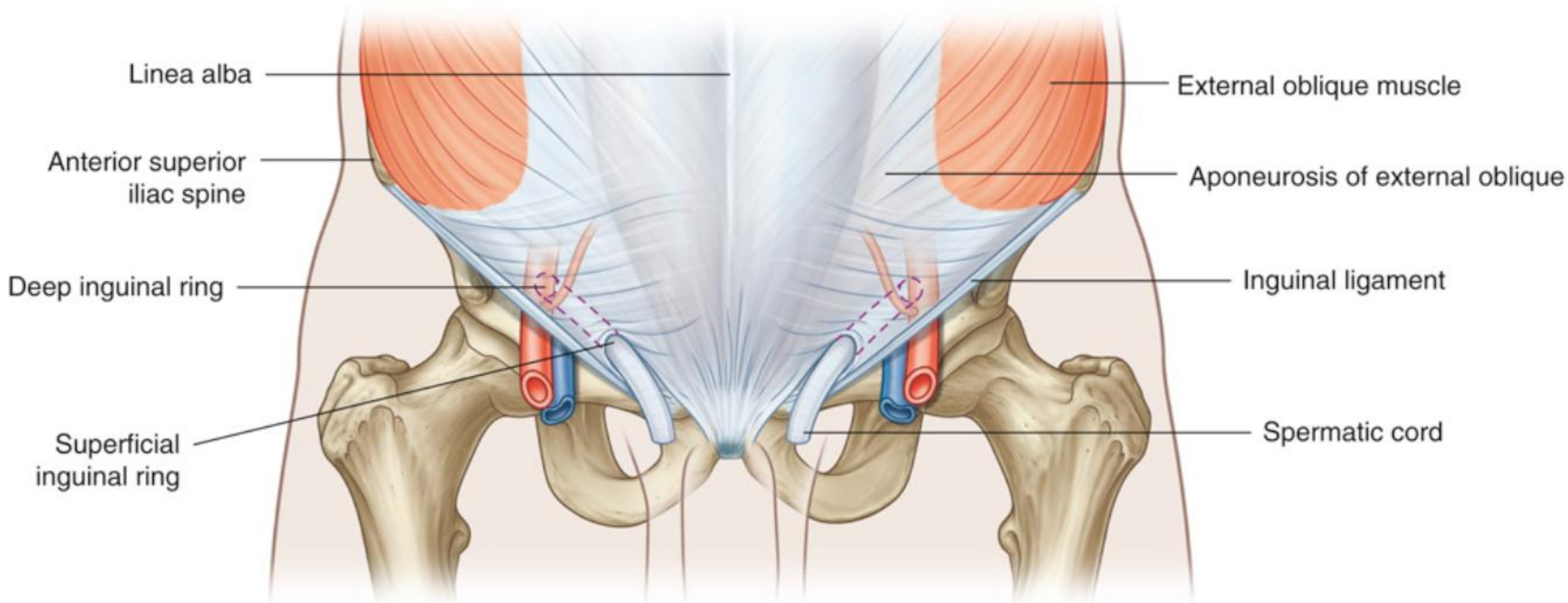
Developmental basis:

- **Processus vaginalis** (peritoneal outpouching) passes through abdominal wall, picking up coverings from abdominal wall → forming the **inguinal canal**.
- Processus vaginalis normally obliterates
- Failure of **processus vaginalis** to obliterate = potential weakness → hernia risk.



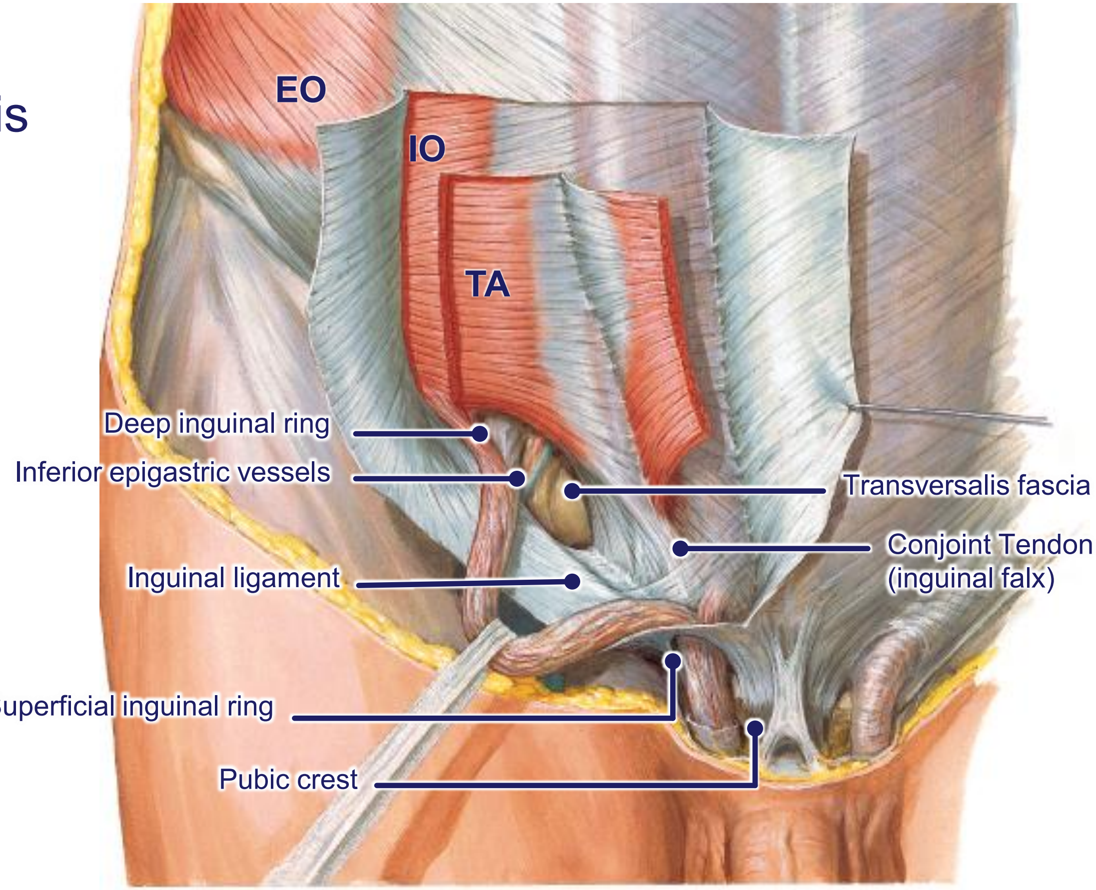
Inguinal canal

- ~4 cm long, oblique passage above & parallel to **inguinal ligament**.
- **Course**: from **deep inguinal ring → superficial inguinal ring**.
- **Deep inguinal ring** (entrance to the canal):
  - Midway between **ASIS** & **pubic symphysis**.
  - Lateral to inferior epigastric vessels.
- **Superficial inguinal ring** (exit from the canal):
  - Above **pubic tubercle**.
  - **Opening** in external oblique aponeurosis.



Inguinal canal – boundaries & contents

- **Anterior wall**: external oblique aponeurosis;
- **Posterior wall**: transversalis fascia
- **Roof**: arching fibers of internal oblique & transversus abdominis converging into **conjoint tendon**.
- **Floor**: inguinal ligament, lacunar ligament.



Main contents:

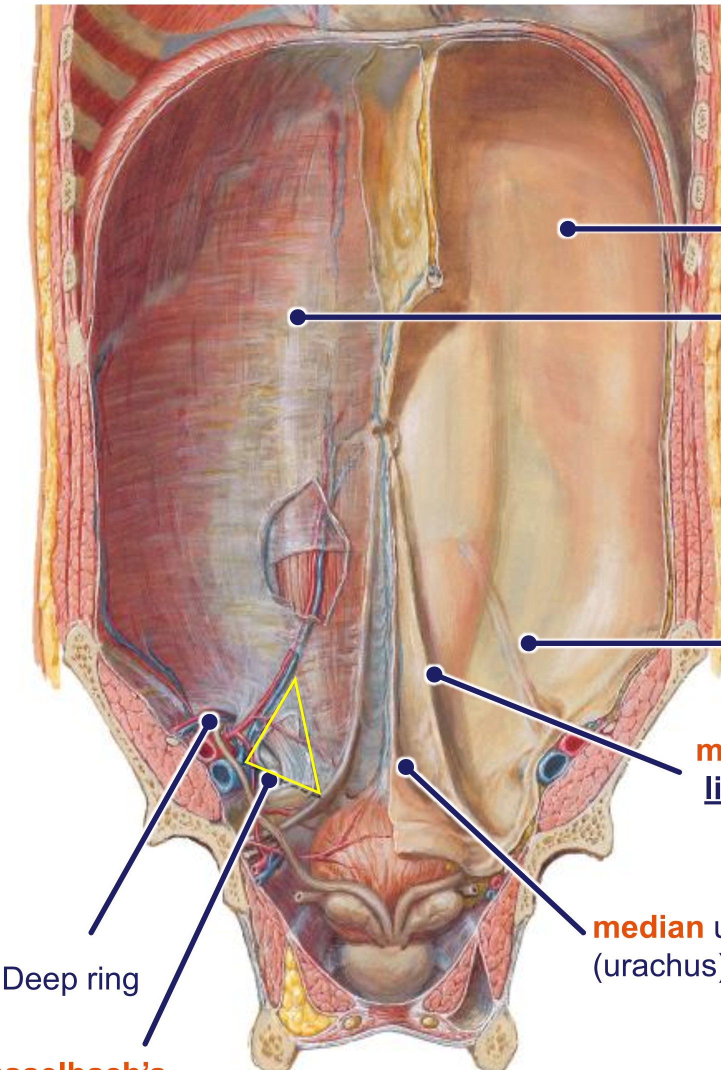
- **Women**: round ligament of uterus + genital branch of genitofemoral nerve) + ilioinguinal nerve (L1)
- **Men**: **spermatic cord** (part of which is genital branch of genitofemoral nerve) + ilioinguinal nerve (L1)

Spermatic cord contents

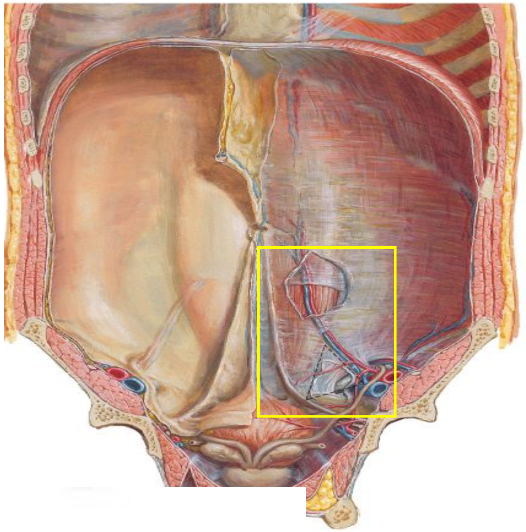
	Male (Spermatic Cord)
Main Structure	Ductus deferens (Vas deferens)
Arteries	- Testicular artery - Artery to ductus deferens - Cremasteric artery
Veins	Pampiniform plexus → testicular veins
Nerves	- Genital branch of genitofemoral nerve (cremasteric reflex) - sympathetic and visceral afferent nerve fibers,
Lymphatics	Testicular lymphatics → **lumbar (para-aortic) nodes**
Fascial Layers (coverings from abdominal wall) **Mnemonic ICE TIE**	**Internal spermatic fascia** (from **t**ransversalis fascia) **Cremasteric fascia + muscle** (from **i**nternal oblique) **External spermatic fascia** (from **e**xternal oblique aponeurosis)
Peritoneum	- remnants of the processus vaginalis
Exit at Superficial Ring	Continues into **scrotum** (enclosing testis & cord structures)

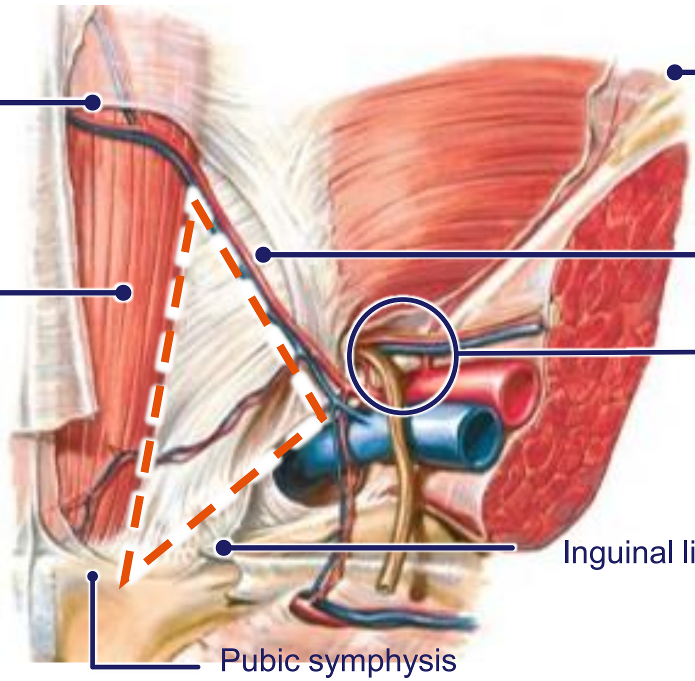
Hasselbach's Triangle and deep inguinal ring

- Inguinal (Hesselbach's) triangle is a region of weakness in the lower anterior abdominal wall
- **Boundaries:**
  - **Medial:** Lateral border of rectus abdominis muscle
  - **Lateral:** Inferior epigastric vessels
  - **Inferior (base):** Inguinal ligament
- **Floor (posterior wall):**
  - Mainly the **transversalis fascia**, lined by **peritoneum.**
- **Nearby landmarks:**
- Overlying peritoneal folds mark embryological remnants:
  - **Median umbilical ligament** (urachus remnant)
  - **Medial umbilical ligaments** (obliterated umbilical arteries)
  - **Lateral umbilical fold** (inferior epigastric vessels)



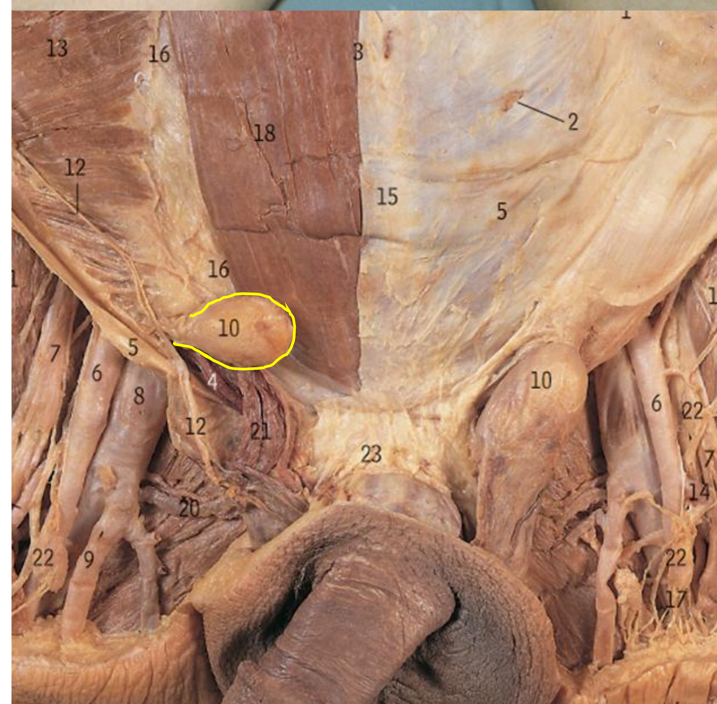
Hasselbach's Triangle and deep inguinal ring





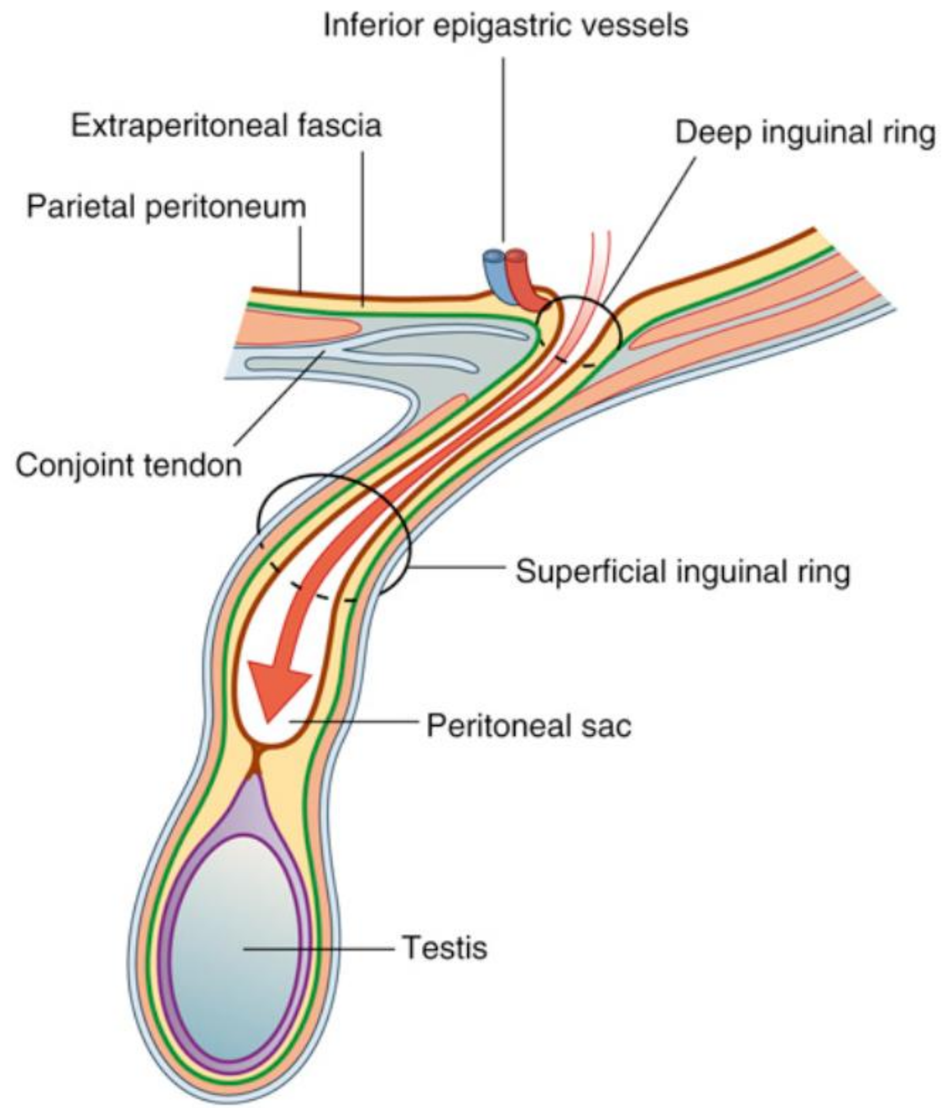
Hernias

- Protrusion of a peritoneal sac (± abdominal contents) through an abnormal or normal opening in the walls that normally contain it.
- Commonest type is **inguinal hernia**
- **Cardinal Features:**
  - **Cough Impulse** – When the patient is asked to cough, the hernia **bulges outward.**
  - **Reducibility** – The swelling can be **pushed back into the abdomen.**
- **Complications:**
  - **Incarceration:** Hernial contents trapped → irreducible, painful swelling → bowel obstruction → urgent surgery.
  - **Strangulation:** Blood supply cut off at neck → ischemia, necrosis, risk of perforation and sepsis → surgical emergency.



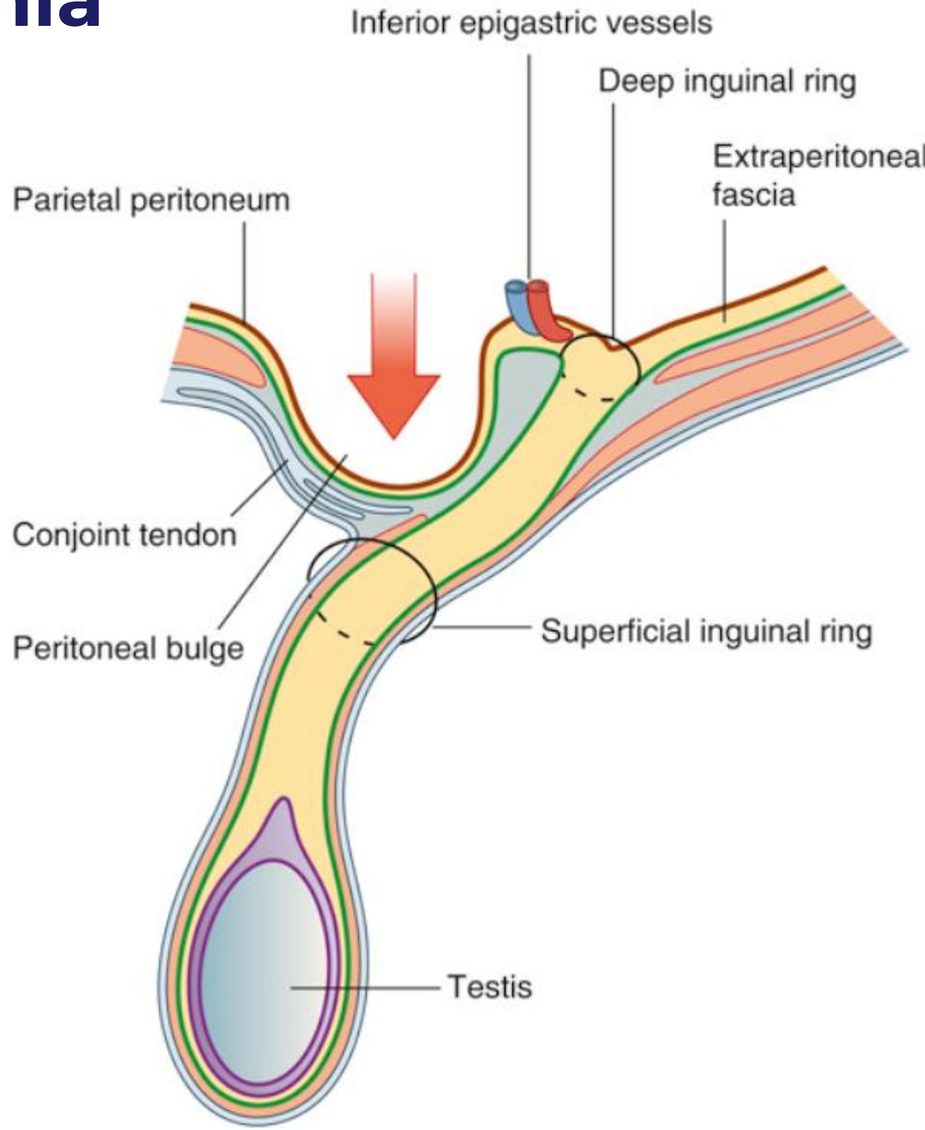
Indirect Inguinal hernia

- **Most common** type; more frequent in men.
- **Congenital** → persistence of **patent processus vaginalis**.
- **Path**: Enters canal via **deep inguinal ring** (lateral to inferior epigastric vessels) → pass through canal → superficial inguinal ring → **scrotum (men)** or **labia majus (women)**.



Direct Inguinal hernia

- **Acquired** type; seen in older men with weakened abdominal muscles (age, obesity, heavy lifting, chronic coughing or straining)
- **Path**: Protrudes directly through **posterior wall of inguinal canal** → Bulges **medial to inferior epigastric vessels** in **Hesselbach's triangle** → exit via superficial inguinal ring.
- Usually does **not traverse full length of canal**.



Inguinal hernias and femoral hernia

	Indirect Inguinal Hernia	Direct Inguinal Hernia	Femoral Hernia
Origin	Congenital (patent processus vaginalis)	Acquired (weak abdominal wall, aging, chronic cough, surgery)	Acquired (weakness at femoral canal, increased intra-abdominal pressure)
Age Group	Young men	Older men	More common in women
Pathway	Enters **deep inguinal ring** → through canal → exit superficial ring	Pushes directly through **posterior wall of canal** → exit superficial ring	Protrudes through **femoral canal** below inguinal ligament
Relation to inferior epigastric artery	**Lateral**	**Medial**	**-**
Relation to inguinal ligament	**above** inguinal ligament (above pubic tubercle)	**above** inguinal ligament (above pubic tubercle)	**Below** inguinal ligament**,** (inferolateral to pubic tubercle)
Extent	May travel entire canal; often into **scrotum** (men) or **labia majus** (women)	Limited; bulges forward; rarely extends to scrotum	Rarely extends upward; bulges in **upper thigh/groin crease**
Coverings	Acquires **all 3 cord layers** (internal spermatic fascia, cremasteric fascia, external spermatic fascia)	Only **external spermatic fascia** (if passing superficial ring)	No spermatic cord coverings; surrounded by femoral sheath
Complications	Obstruction, strangulation (esp. in narrow canal)	Less likely to strangulate than indirect	High risk of strangulation due to narrow femoral ring

Use of Transillumination

Hydrocele

Collection of fluid in the tunica vaginalis

Usually in infants

Can be communicating or non-communicating depending on the processes vaginalis being completely obliterated or not

Transillumination may reveal it is fluid

Indirect hernia

Congenital

Does not transilluminate

Abdominal incisions: Midline and others

Determined by several factors

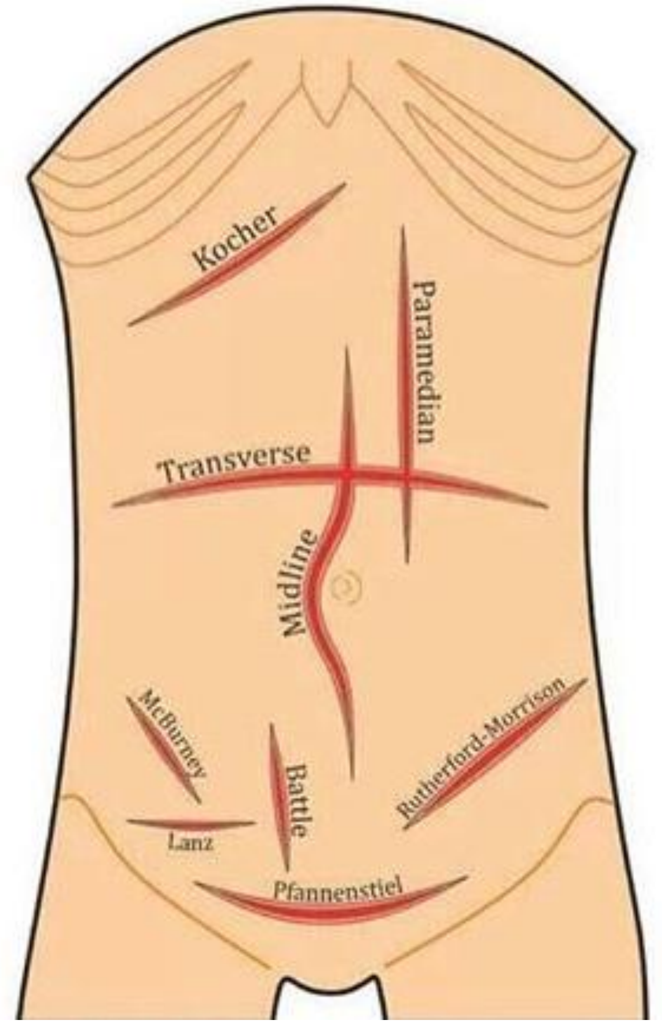
- Organ(s) of interest
- Surrounding structures
- Location of blood vessels and nerves
- Strength of tissue for suture placement
- Clinical information

Considerations

- Access, ability to enlarge site when necessary

Laparoscopic incisions

- Ports are placed through small incisions
- Umbilicus is used for 1° port = camera
- Additional ports are placed at 60° angles and at least 5cm from each other
- Depends highly on area of exploration or procedure



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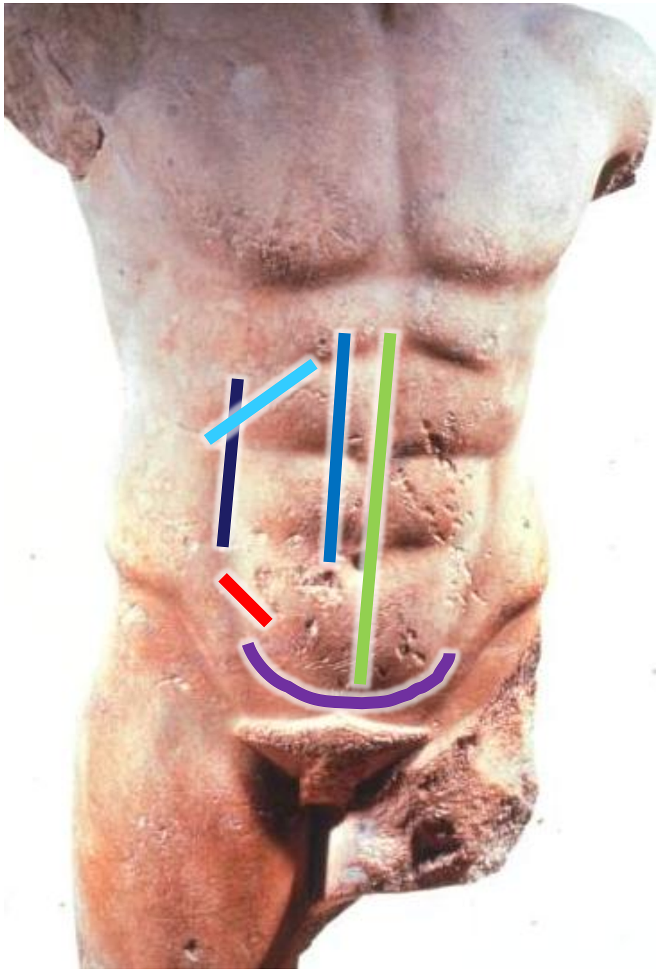
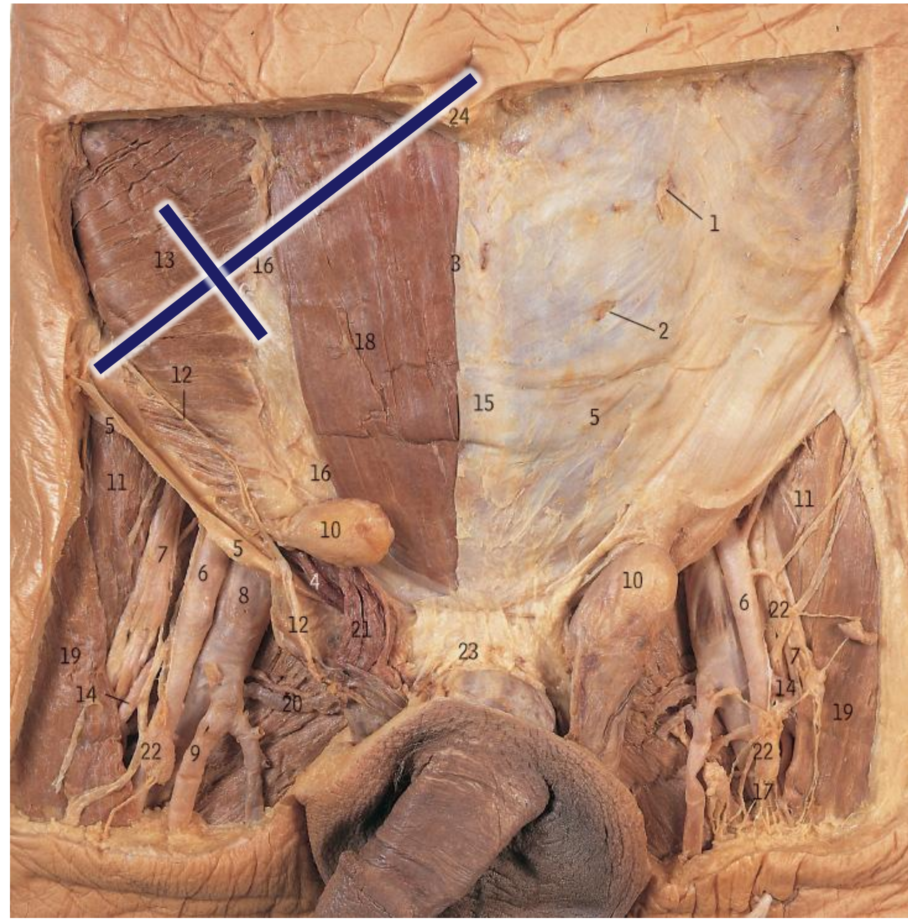
Abdominal Incisions at Linea semilunaris & Costal Margins

- **Median** - At the linea alba good for exploratory laparotomy; advantage is that no blood vessels cross the area, good suture repair
- **Para-median** - Laterally to the linea alba
- **Pfannenstiel** - Above the pubic symphysis (bikini line through external abdominal oblique and rectus abdominis muscles)
    - Caesarean section
- **Mc Burney's point** - 2/3 between the umbilicus and ASIS
    - Appendectomy
- **Linea semilunaris** - Incision is made at linea semilunaris
- **Kocher** - Subcostal
    - For access to the gallbladder

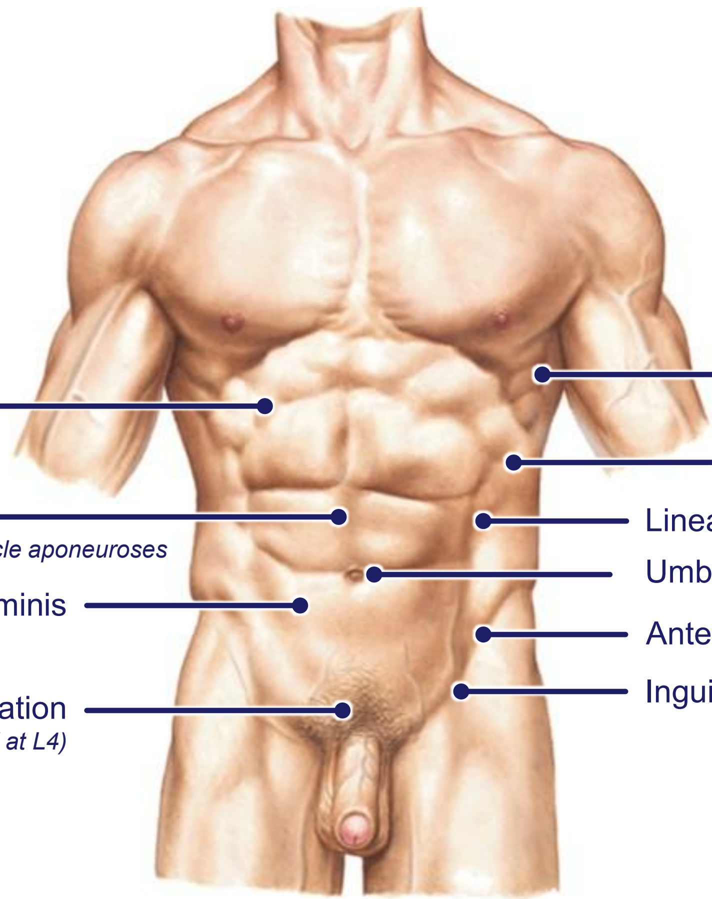


McBurney's point:
Surgical approach and significance

- 2/3 the distance from the umbilicus to anterior superior iliac spine
- Locate the base of the cecum
    - origin of the appendix
- Incision is made through subcutaneous tissue
- The transversalis fascia, extraperitoneal fascia and parietal peritoneum is also cut
- Avoids ilioinguinal nerve (#12) and iliohypogastric nerve



Surface Anatomy



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